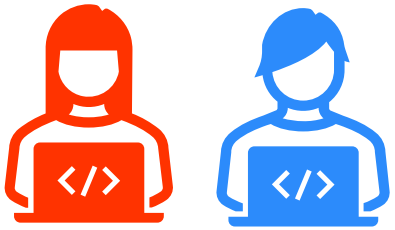
SANDY workshop

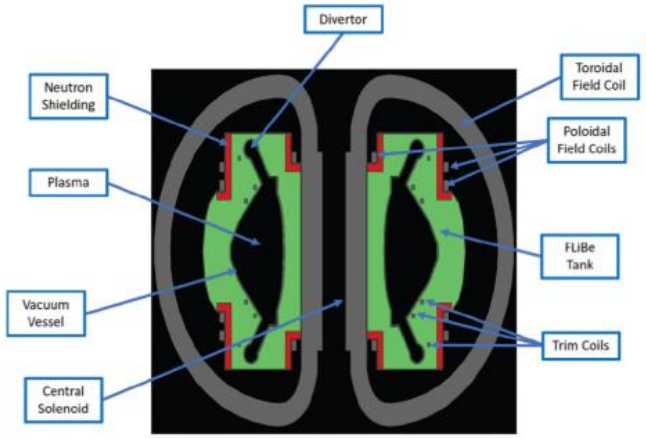
1. Introduction / presentation

2. Hands-on training
   - Installation
   - Exercises

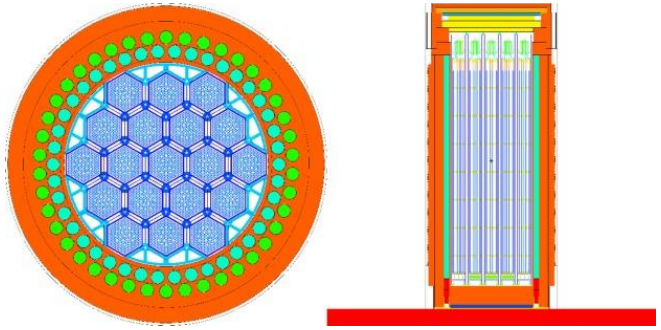


https://github.com/luca-fiorito-11/sandy

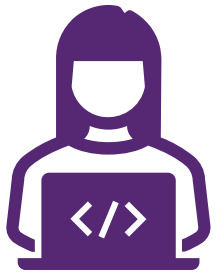
https://doi.org/10.1080/00295639.2022.2153638

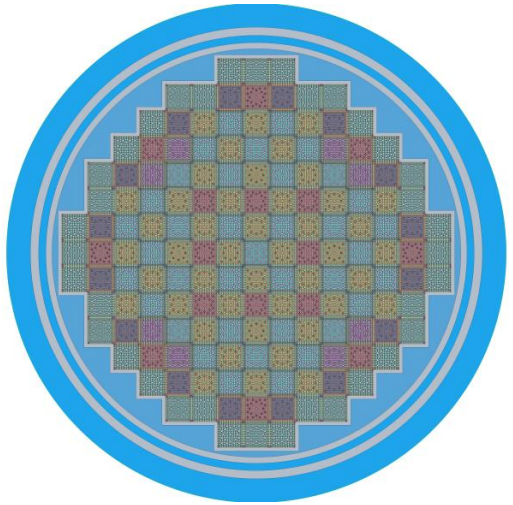


https://doi.org/10.1016/j.nucengdes.2024.113723

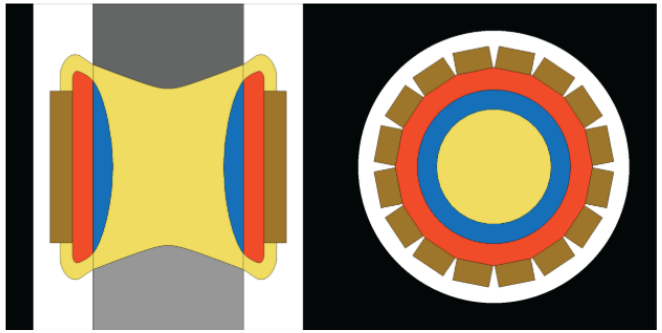


Computer code / Simulation



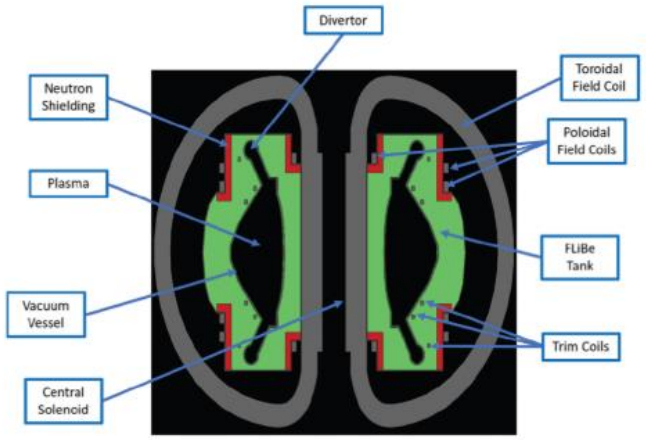


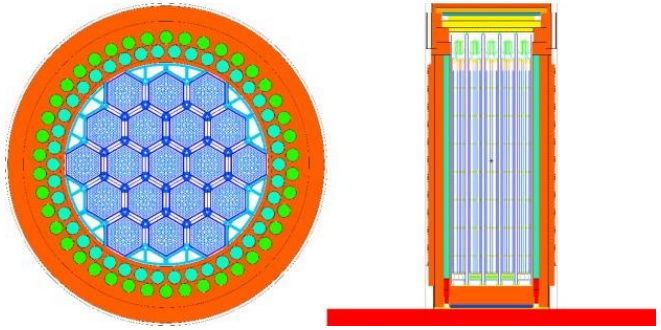
https://publications.polymtl.ca/10545/



(a) *x-z plane* (b) *x-y plane*

https://doi.org/10.1080/00295639.2023.2190861





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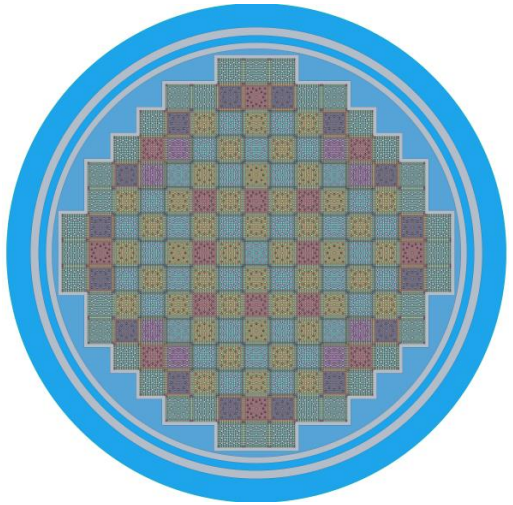
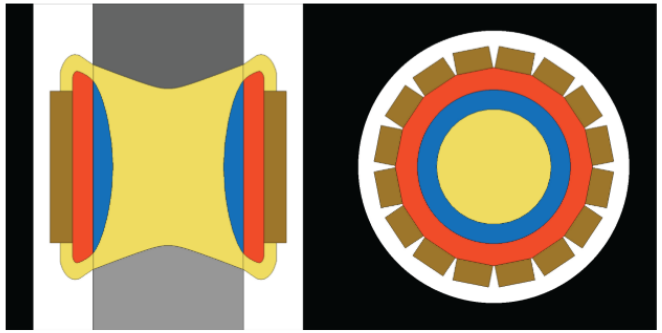
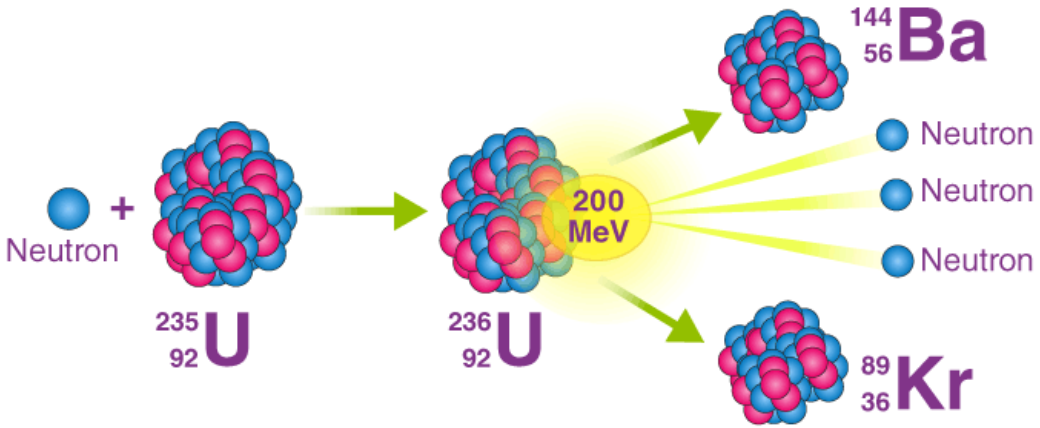




(a) *x-z plane* (b) *x-y plane*

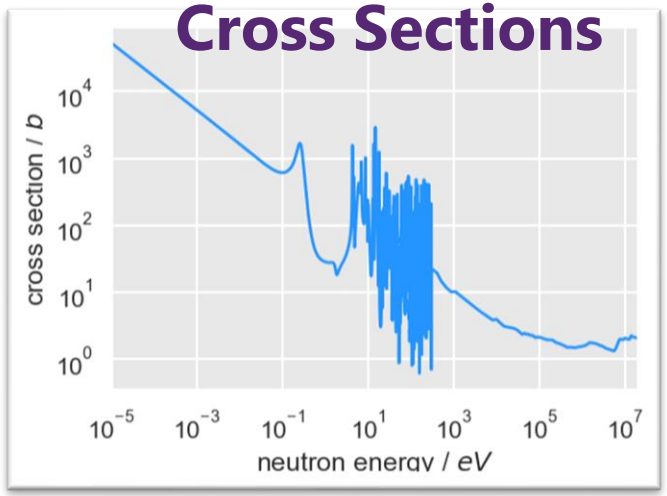
Nuclear Data

What are Nuclear Data?

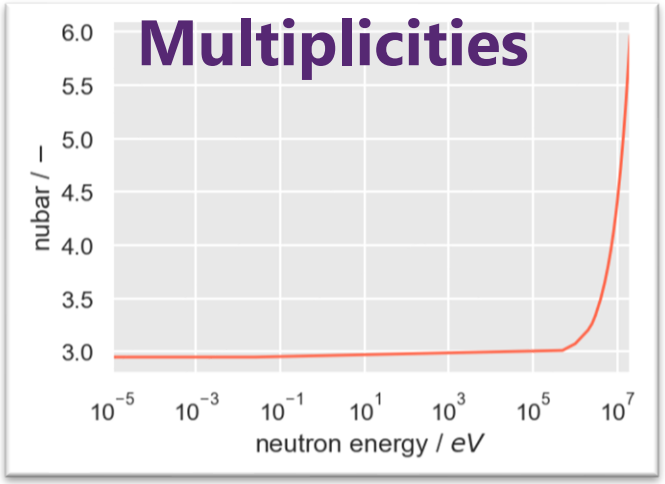


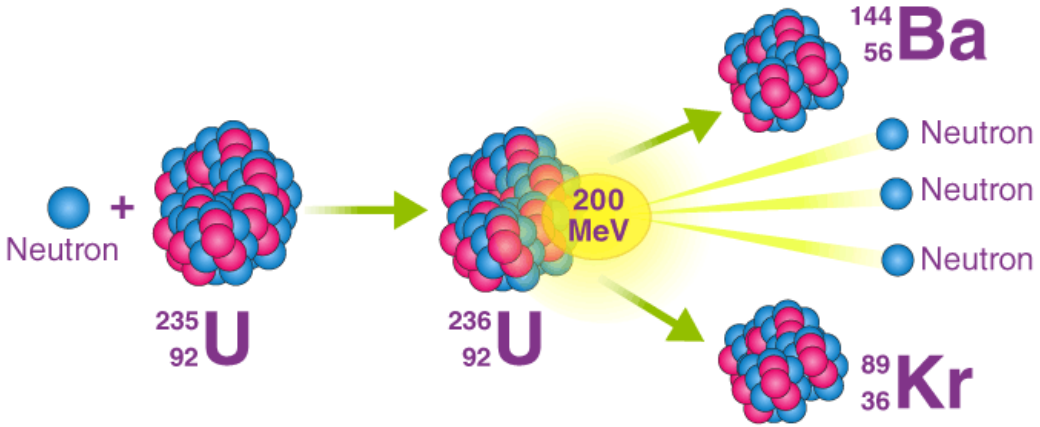
What are Nuclear Data?

Cross Sections

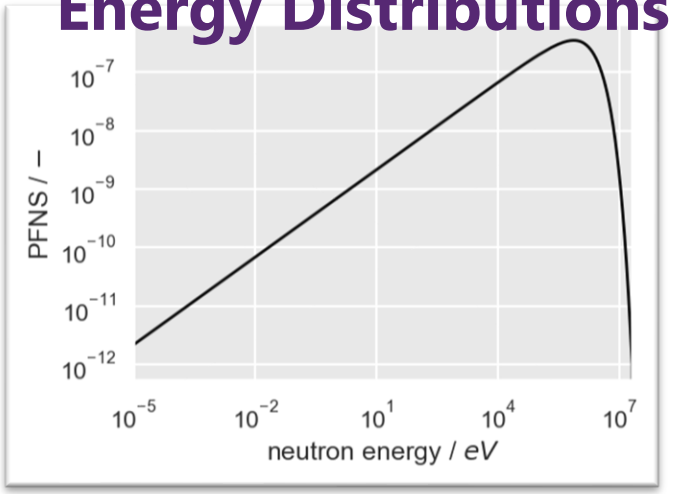


Multiplicities

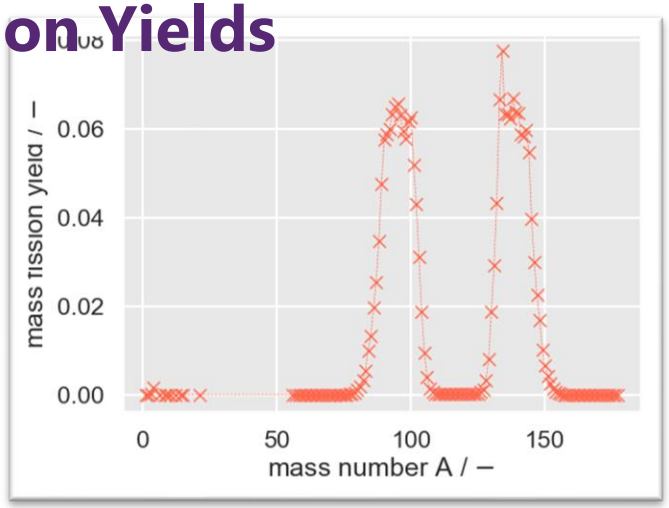




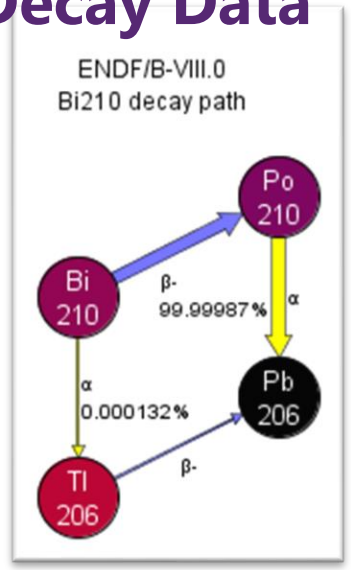
Energy Distributions



Fission Yields



Decay Data



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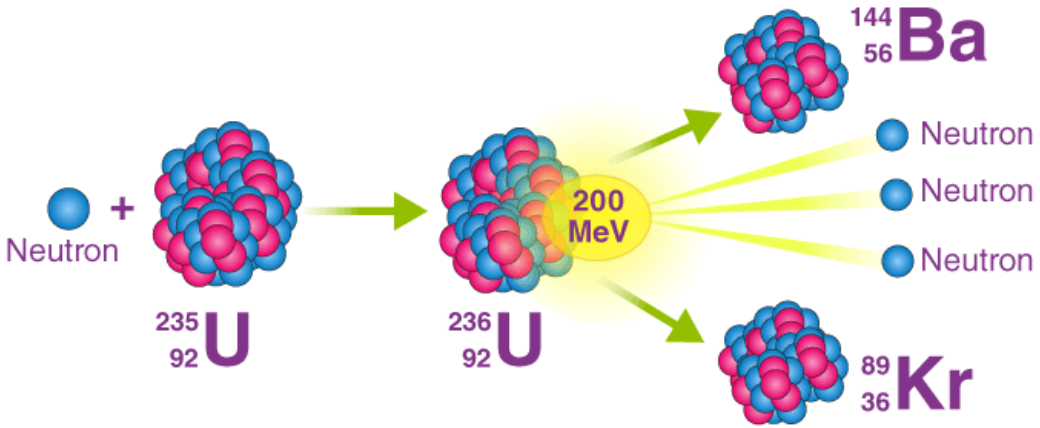

Uncertain !

What is “uncertainty”?

General Concept

The word “uncertainty” means doubt, and thus in its broadest sense “uncertainty of measurement” means doubt about the validity of the result of a measurement.

(JCGM 100:2008 GUM 1995)

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Specific Quantity

Parameter, associated with the result of a measurement, that characterizes the dispersion of the [...] measurand.

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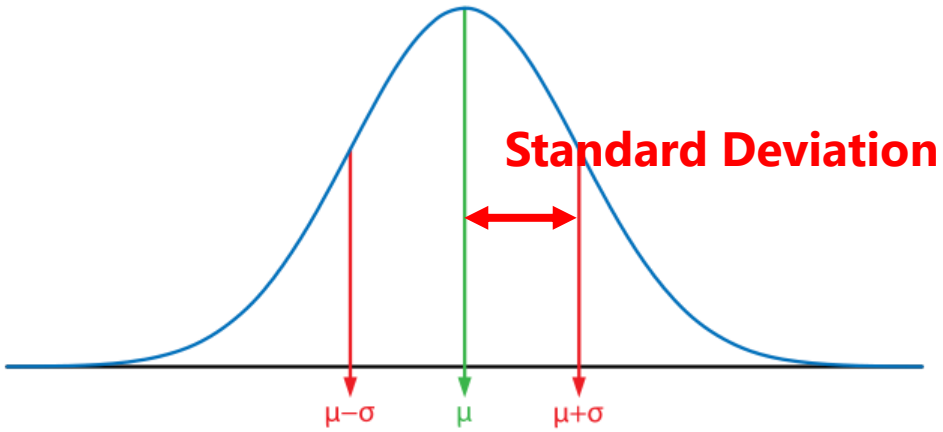

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(JCGM 100:2008 GUM 1995)

“uncertainty”
=
1σ standard deviation

Uncertainty Quantification

- Decision making requires uncertainty!
  - w/o uncertainty : **binary decision**
  - w/ uncertainty: **probabilistic decision**

- Quantifying how uncertain inputs affect outputs:

Nuclear Data
Uncertainty Propagation

< 1975 ...

Conference on Nuclear Cross Sections and Technology (1975)

R.W. Peele, Uncertainties and Correlations in Evaluated Data Sets Induced by Use of Standard Cross Sections, ORNL

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Deterministic UQ methods

1st order derivative

Deterministic UQ methods

- Very efficient (CPU-wise)
- Requires sensitivities
- Local approximation
- **Approximate the physics to make the problem manageable**

< 1975 | 2010's | ...

Conference on Nuclear Cross Sections and Technology (1975)

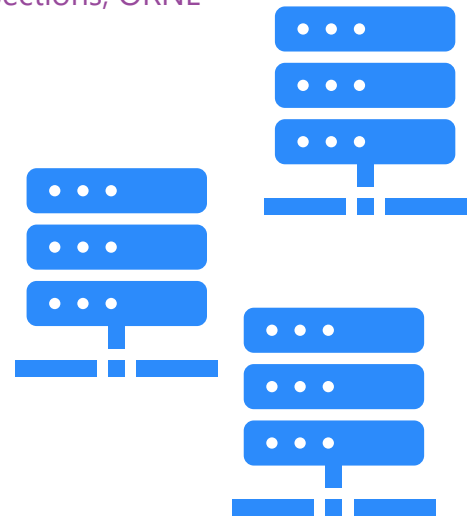
R.W. Peele, Uncertainties and Correlations in Evaluated Data Sets Induced by Use of Standard Cross Sections, ORNL

Deterministic UQ methods

1st order derivative

Stochastic UQ methods

Monte Carlo

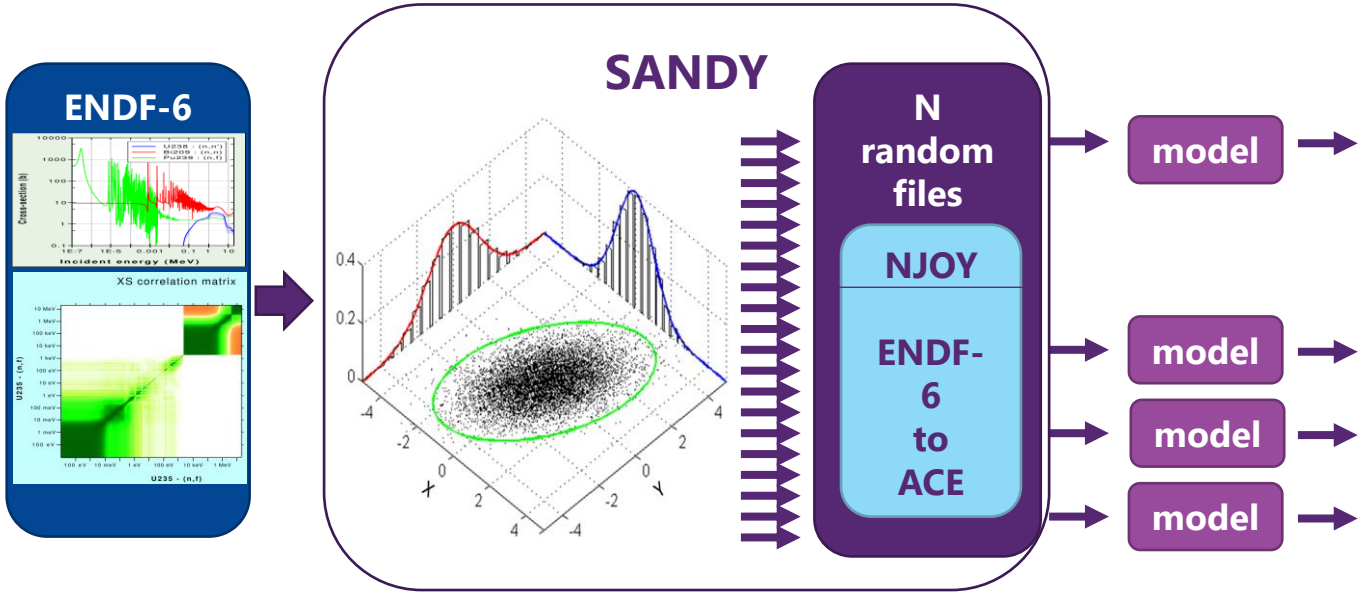


Stochastic UQ methods

- Computationally expensive
- Model treated as a black box
- Global approach
- **Keep the physics exact, and approximate the statistics**

SANDY

Nuclear Data Stochastic Sampling Tool



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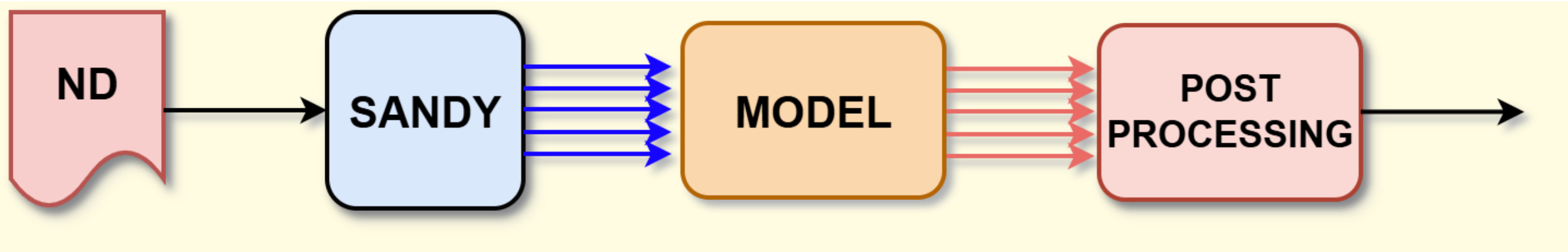
• Decay Data	8

https://github.com/luca-fiorito-11/sandy

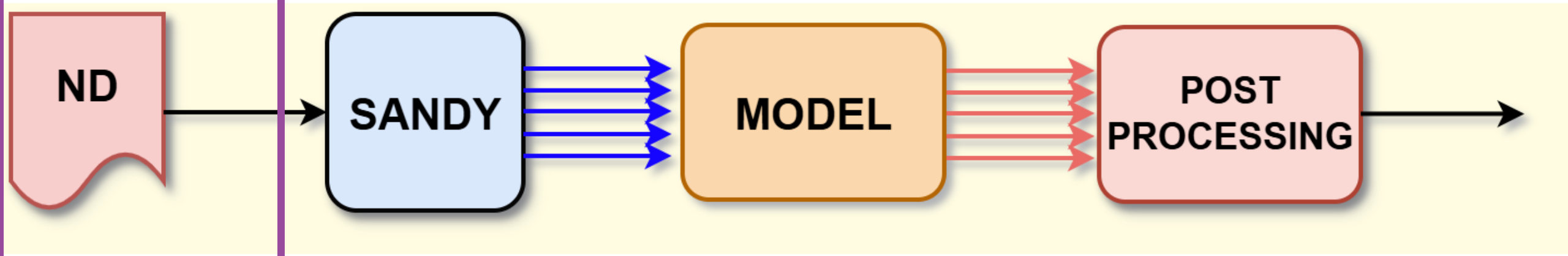
https://luca-fiorito-11.github.io/sandy_notebooks/

Uncertainty Propagation Pipeline



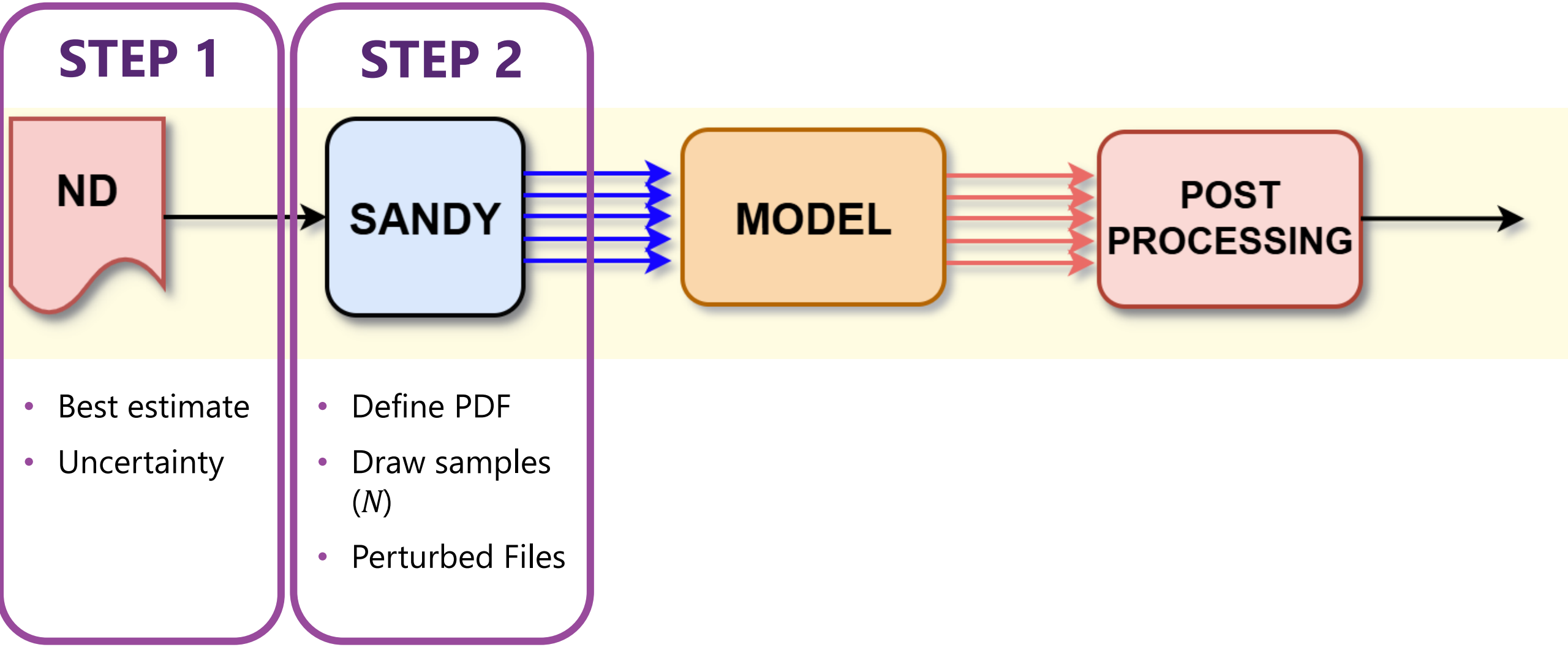
Uncertainty Propagation Pipeline

STEP 1



- Best estimate
- Uncertainty

Uncertainty Propagation Pipeline



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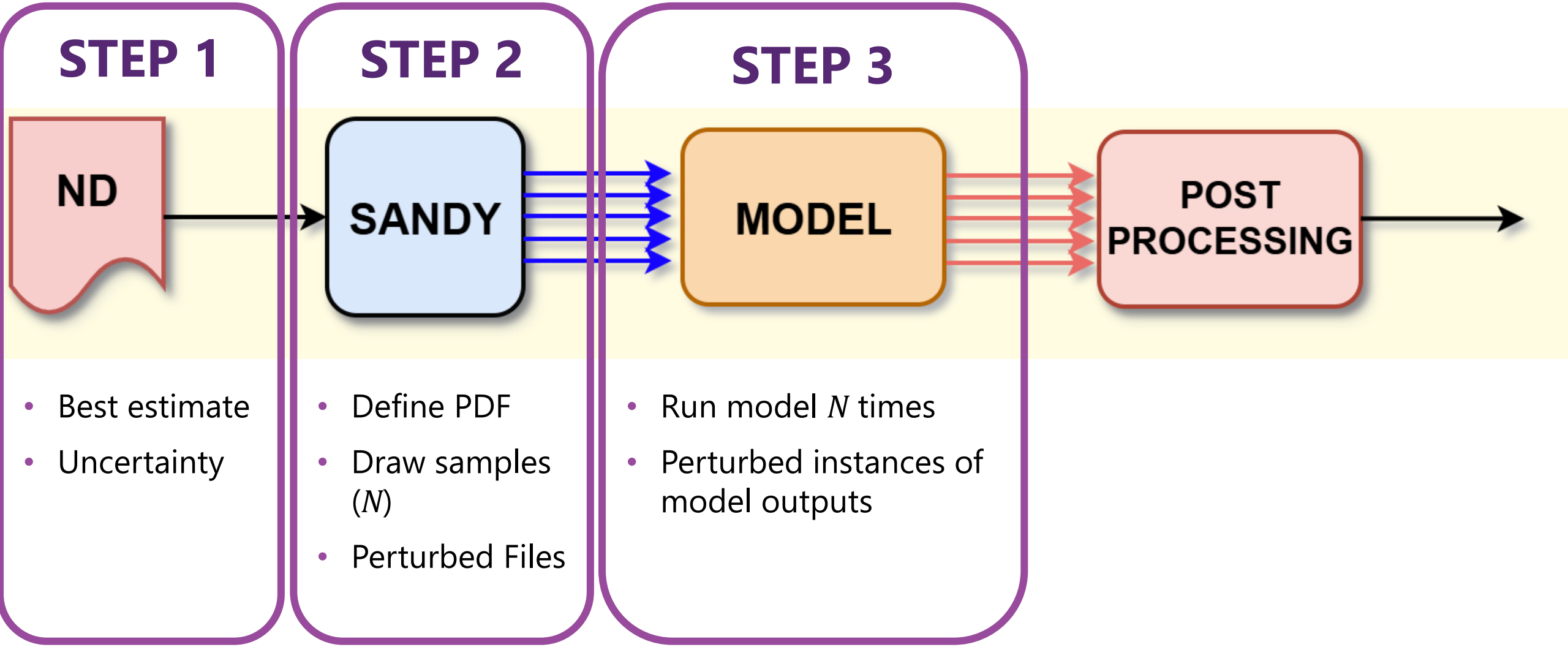

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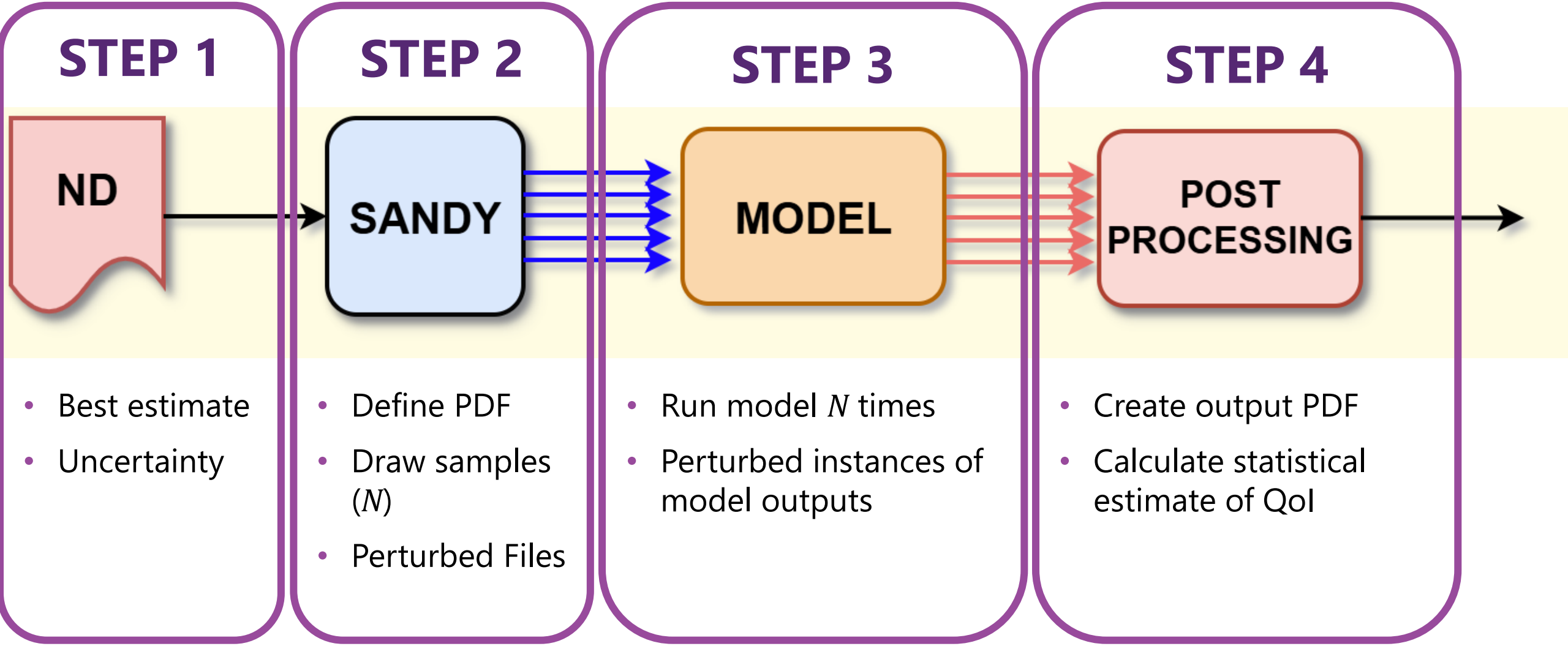
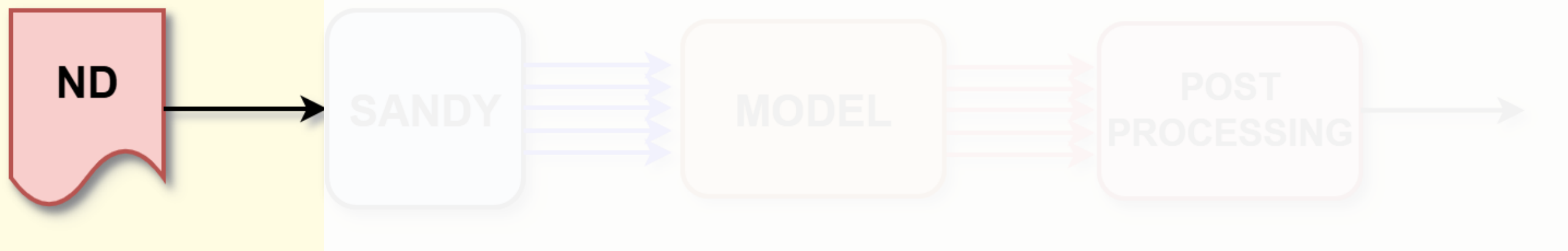
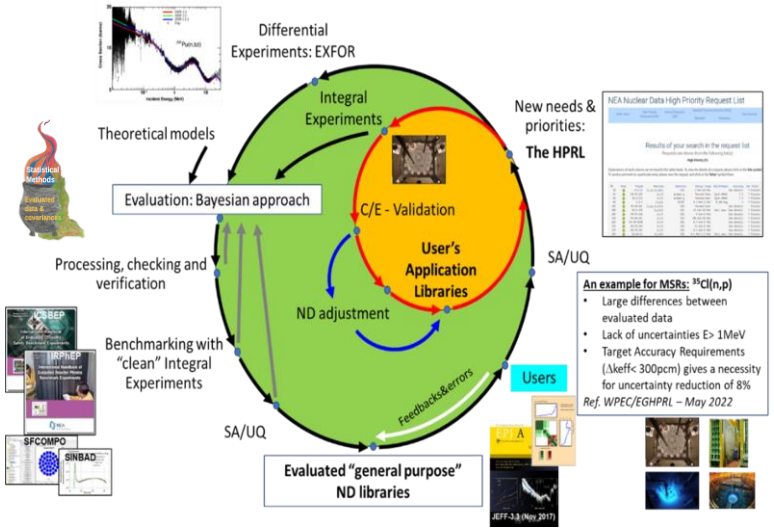




1. Nuclear Data

Nuclear data evaluations

Evaluation process

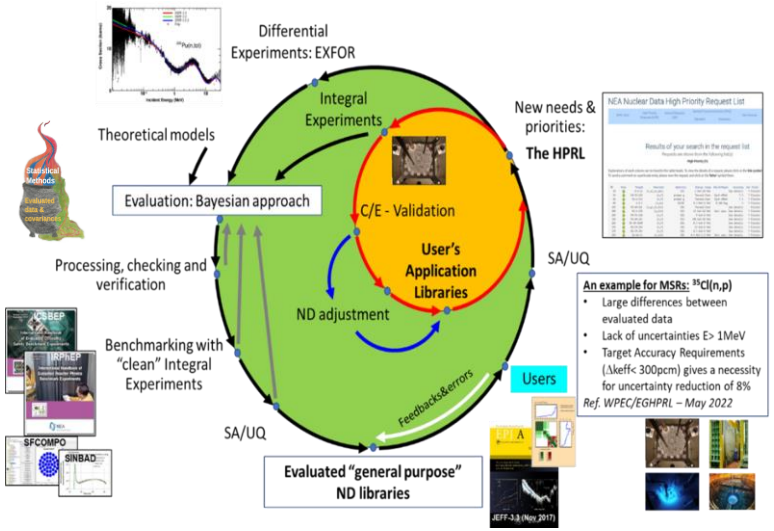


An example of *"nuclear data life cycle"*

Tabulated data

Nuclear data evaluations

Evaluation process



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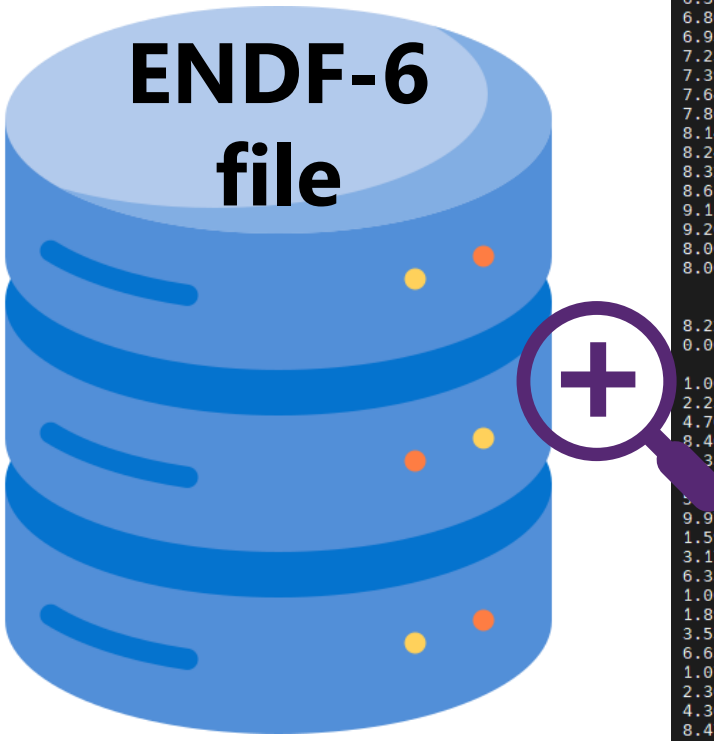

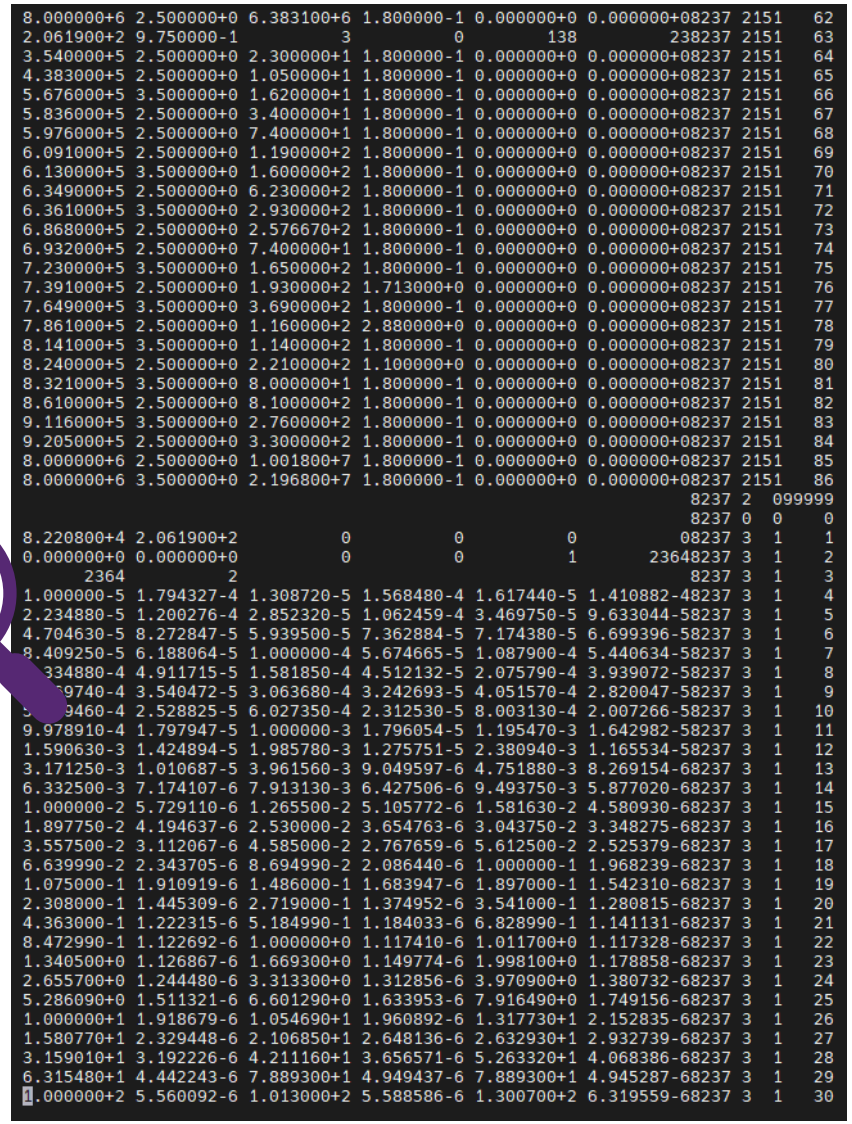

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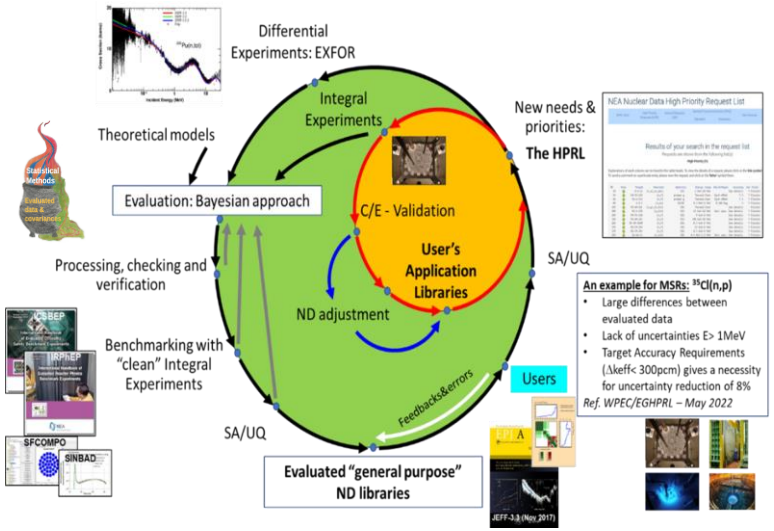
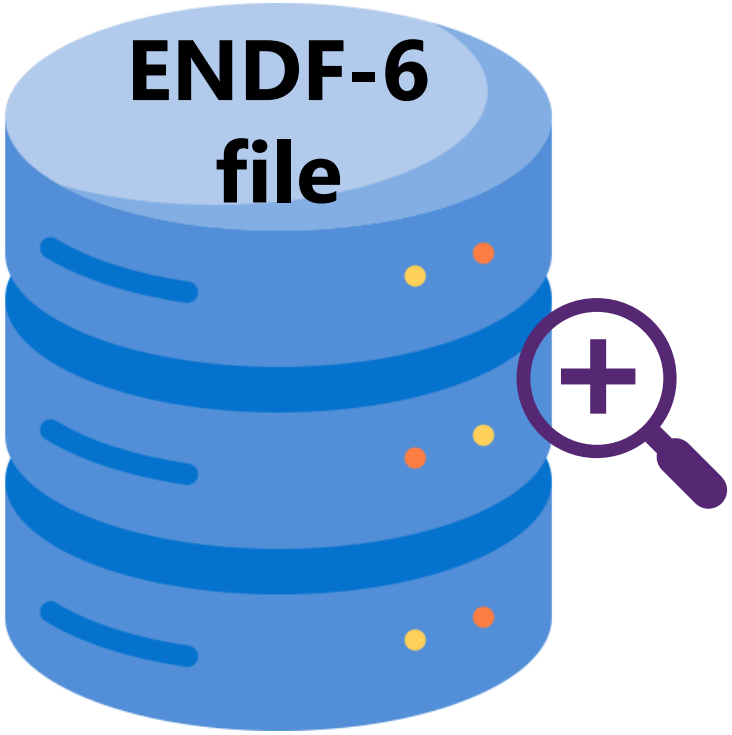
Nuclear data evaluations

Evaluation process



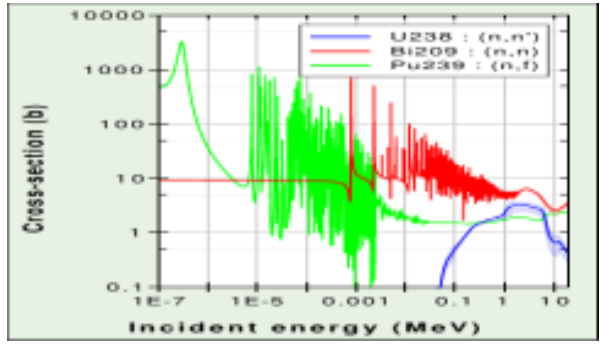
An example of *"nuclear data life cycle"*

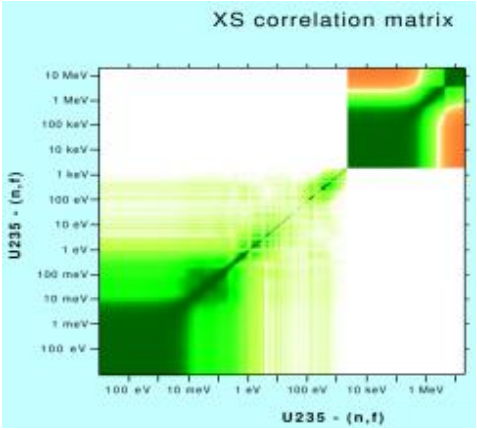
Tabulated data



Python API

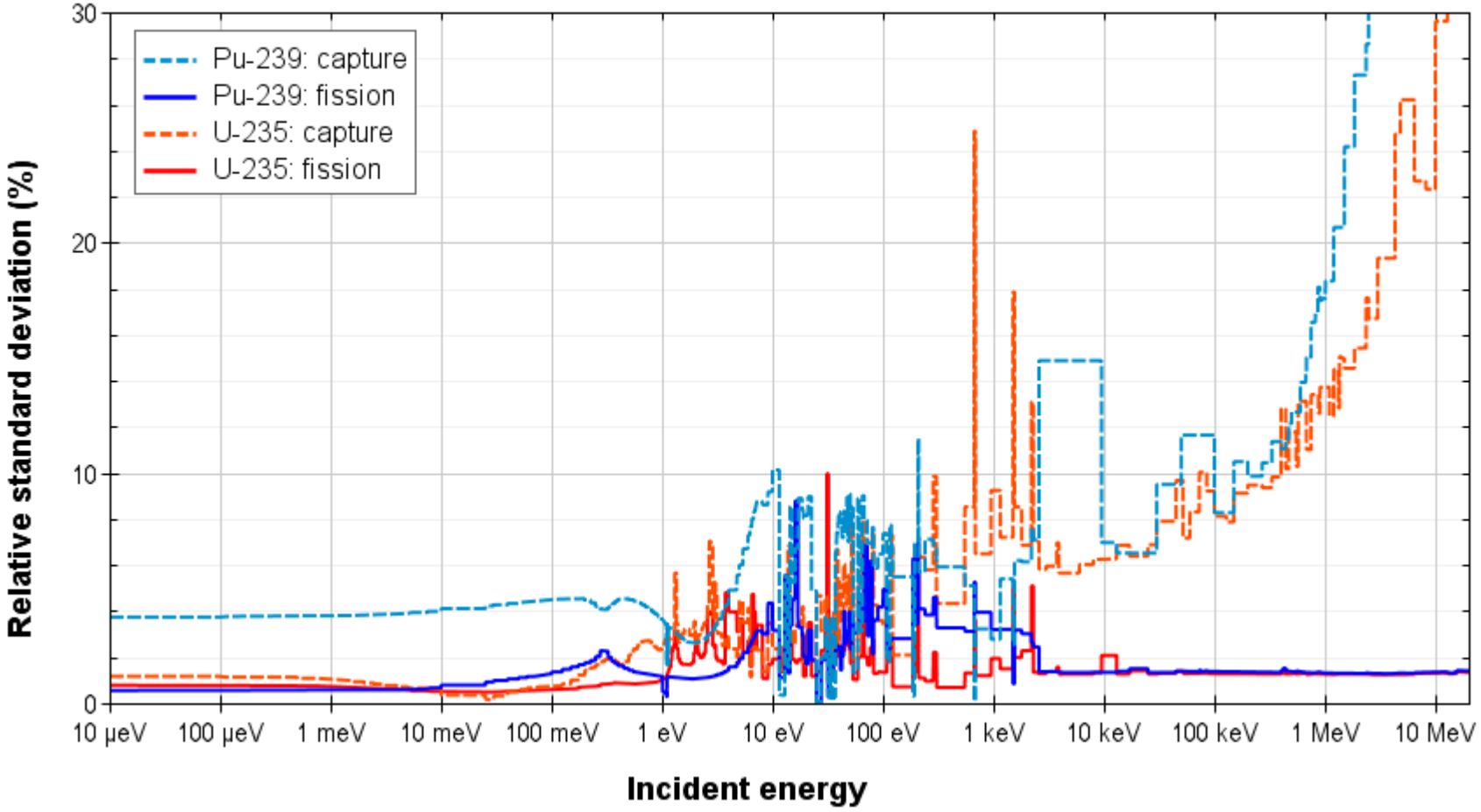
Best estimate



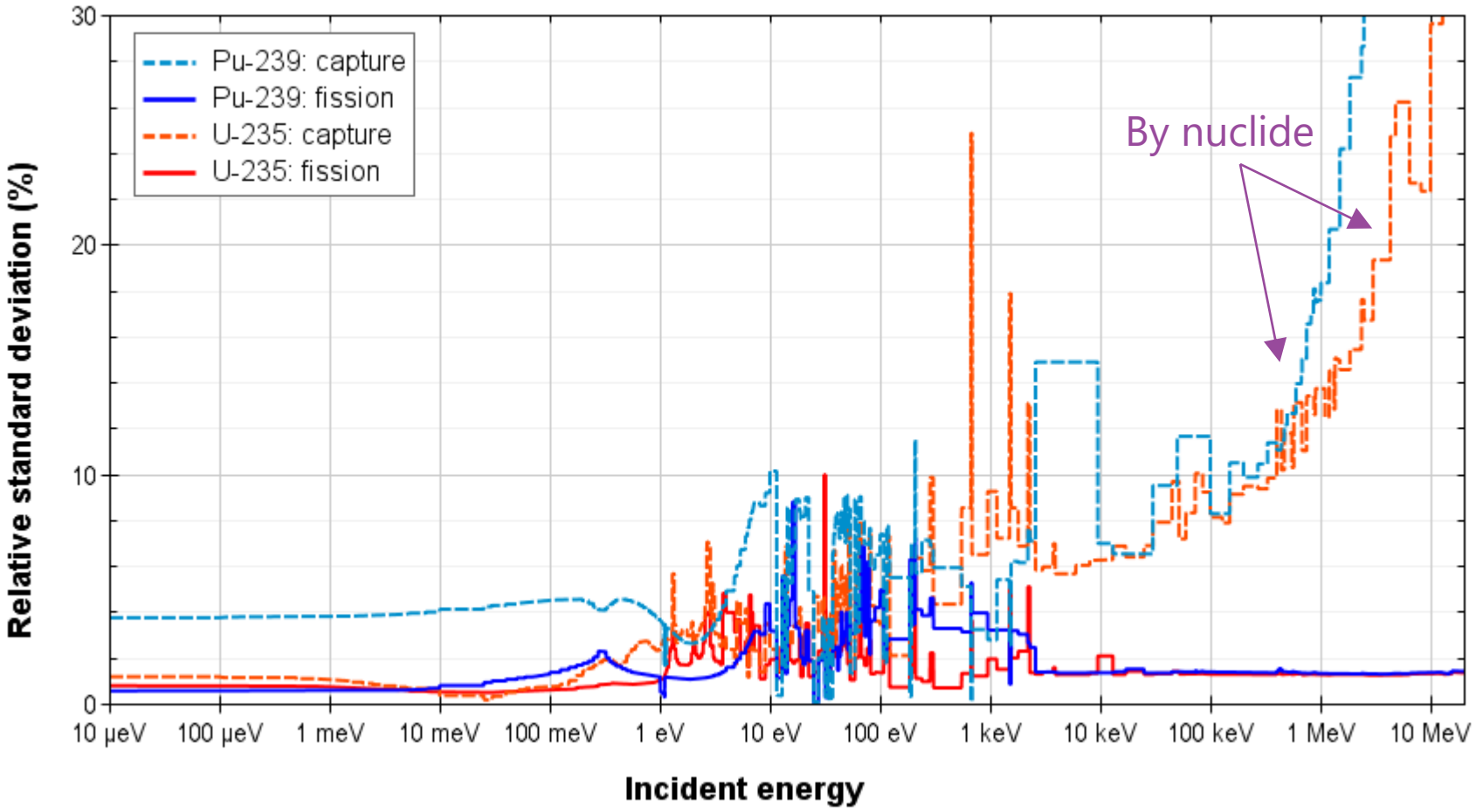


Uncertainty / Covariance matrix

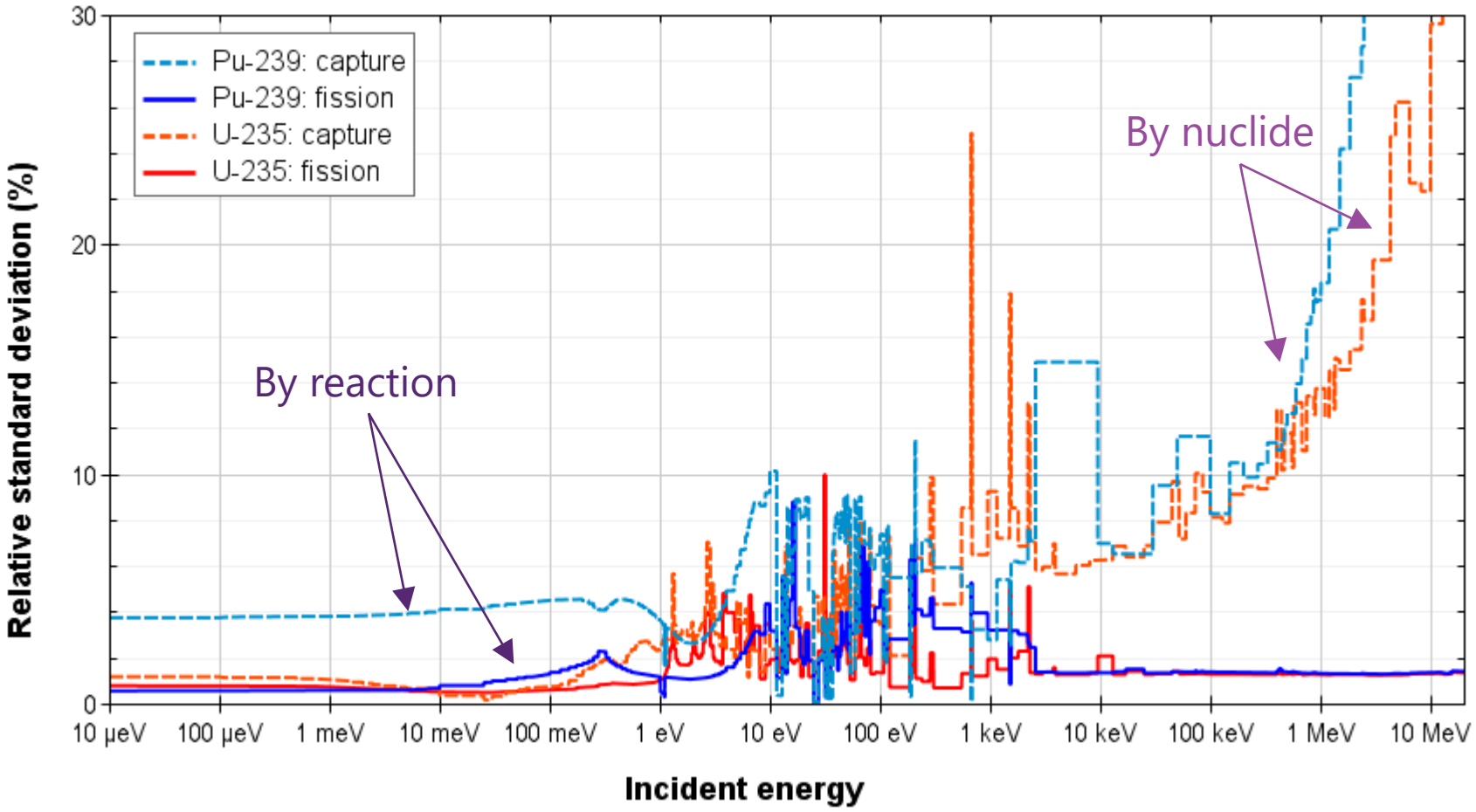
Nuclear data uncertainty



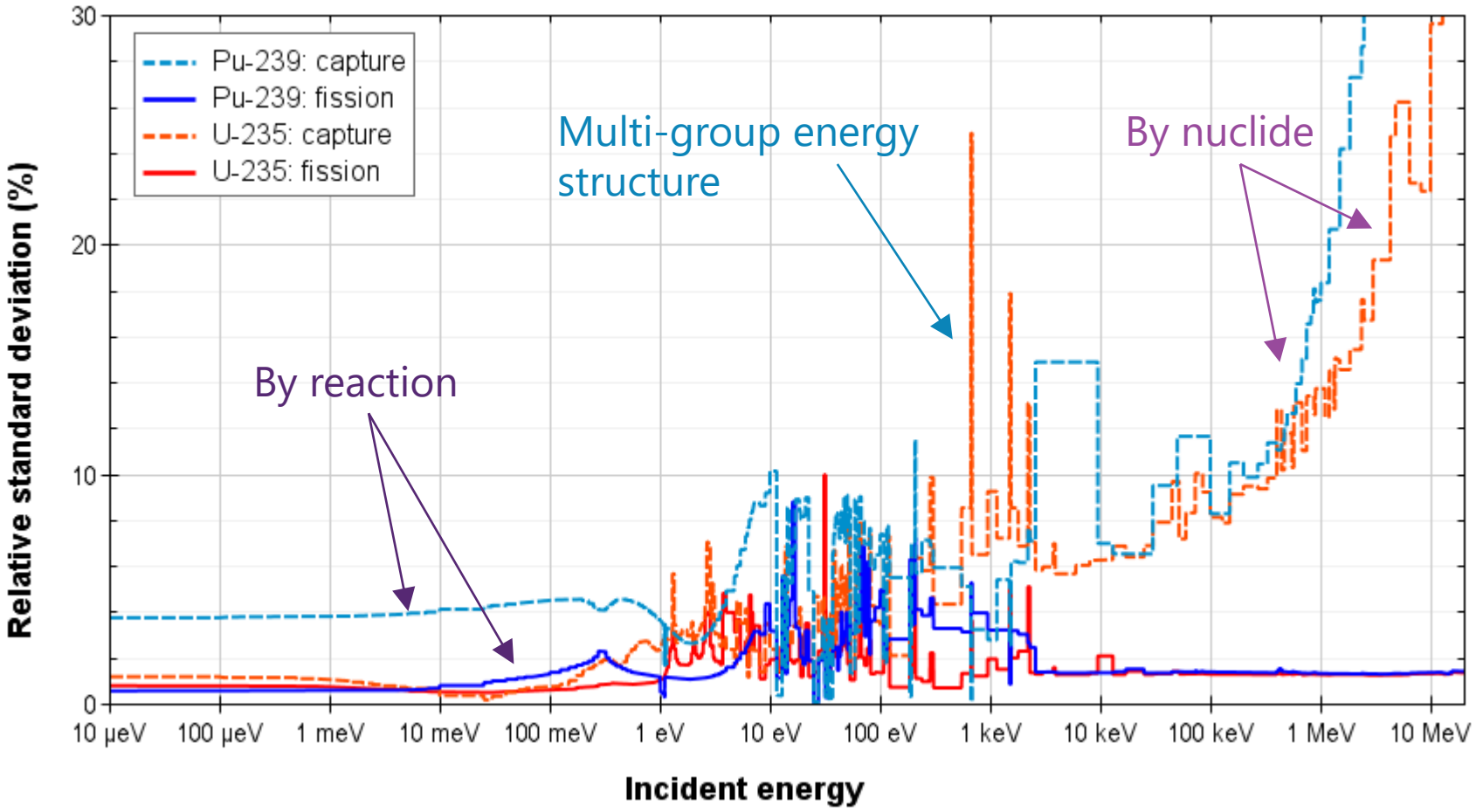
Nuclear data uncertainty



Nuclear data uncertainty



Nuclear data uncertainty

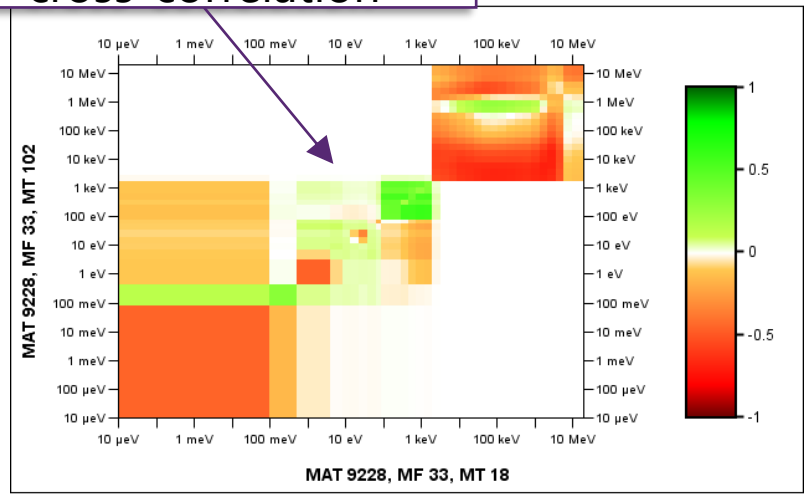


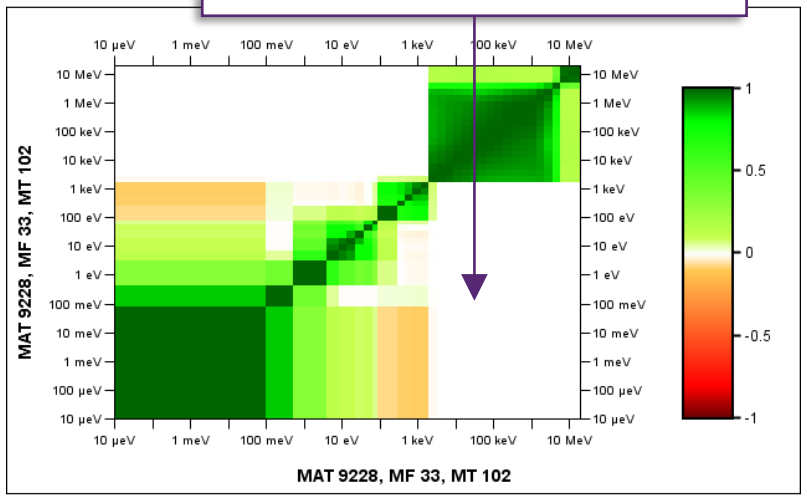
Nuclear data covariance matrix

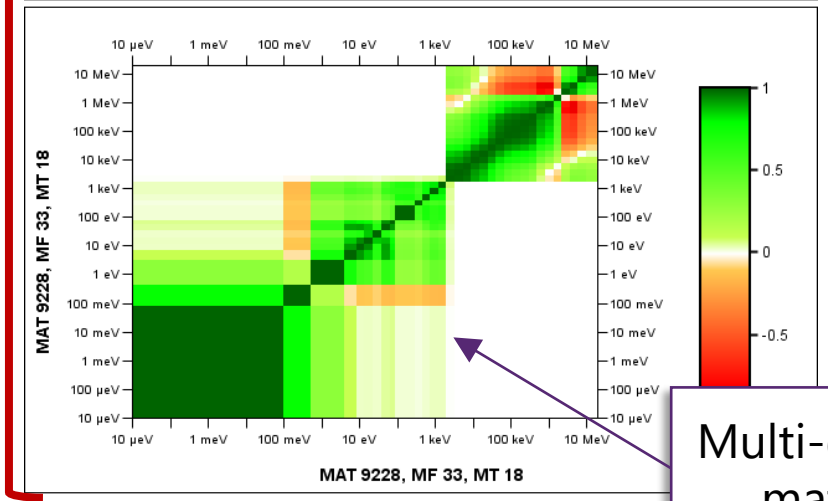
Multi-group **fission-capture** cross-correlation

Multi-group covariance matrix for **capture**

Σ







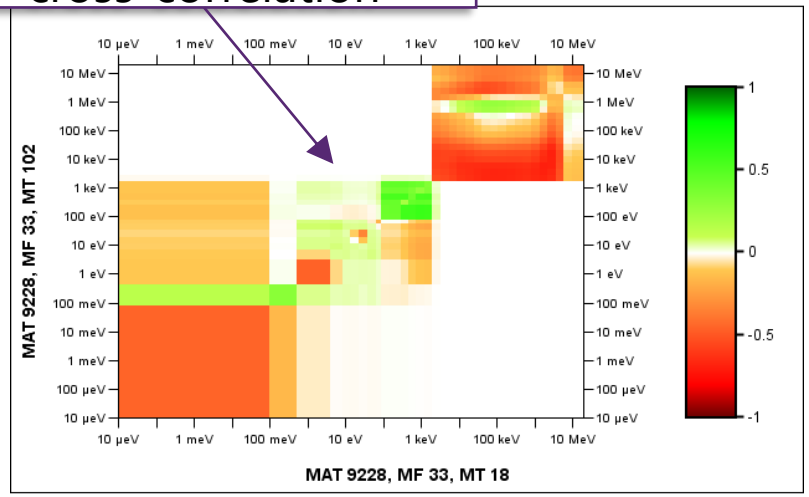
Multi-group covariance matrix for **fission**

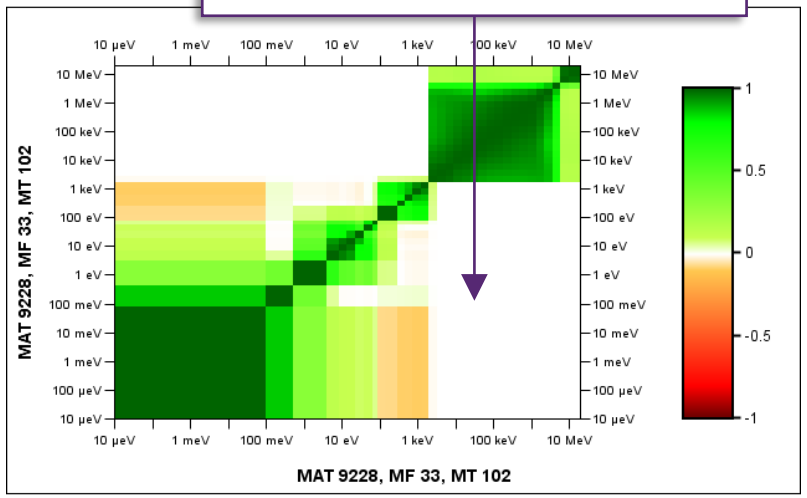
Nuclear data covariance matrix

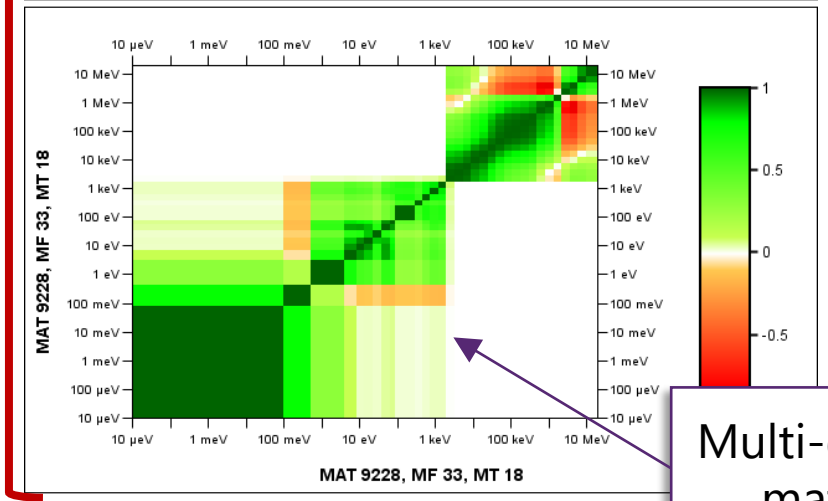
Multi-group **fission-capture** cross-correlation

Multi-group covariance matrix for **capture**

Σ





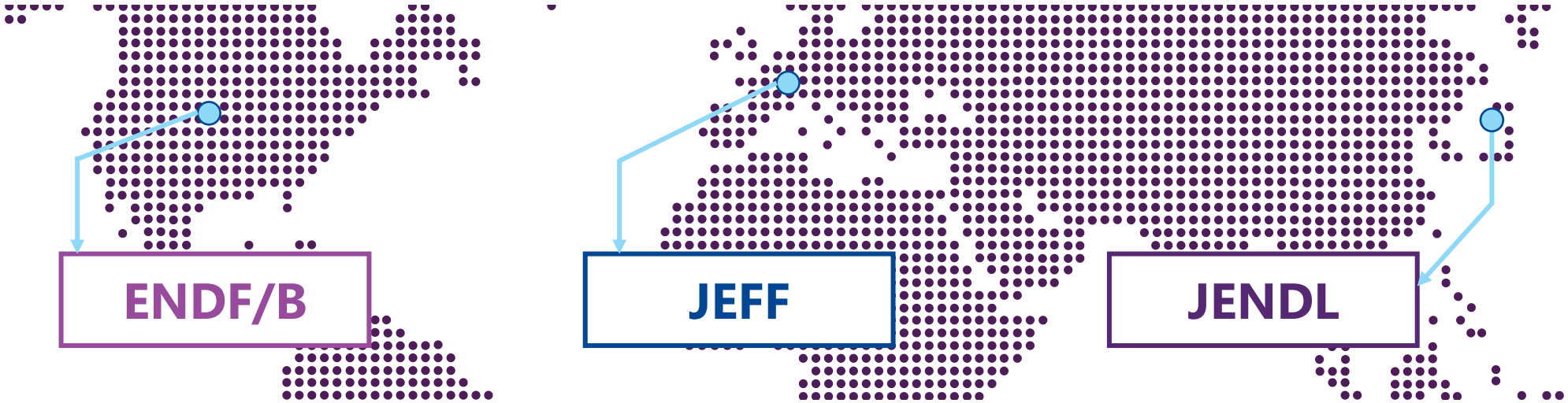


Covariance: measure of the joint variability of two random variables

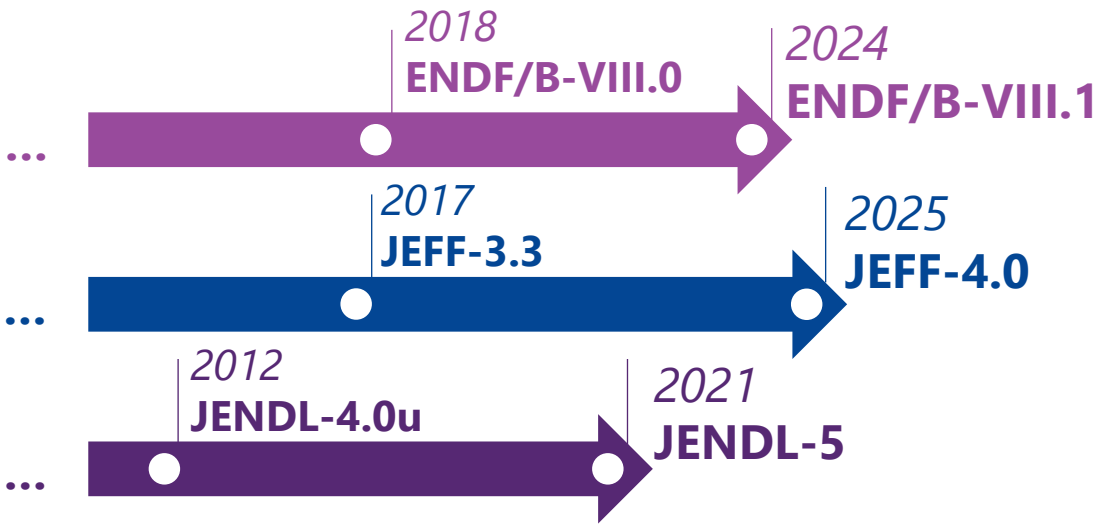
Correlation coefficient: normalized measurement of the covariance, such that the result always has a value between −1 and 1

Multi-group covariance matrix for **fission**

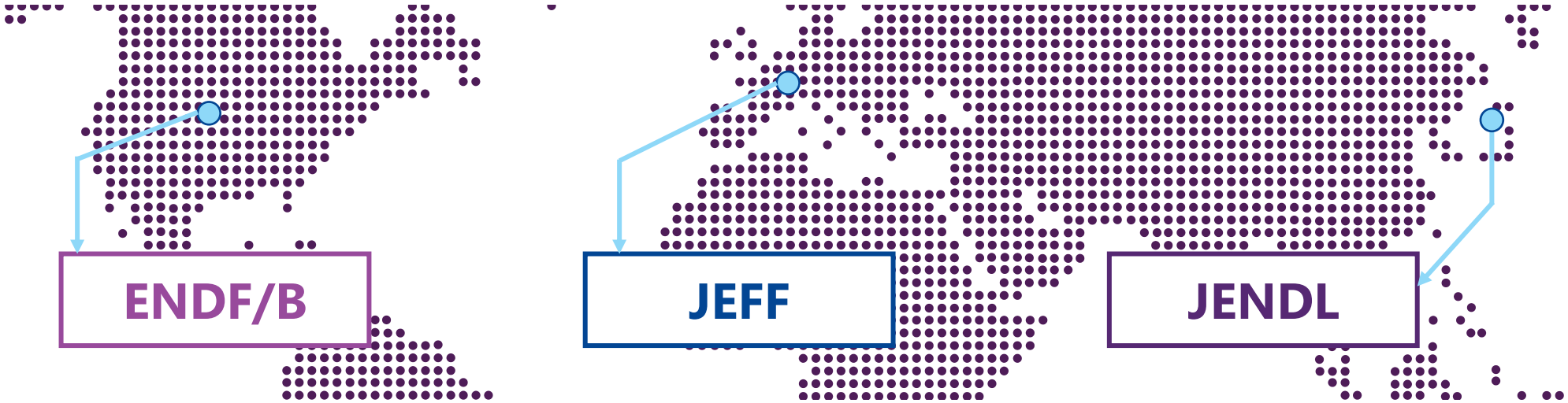
Nuclear data evaluation project



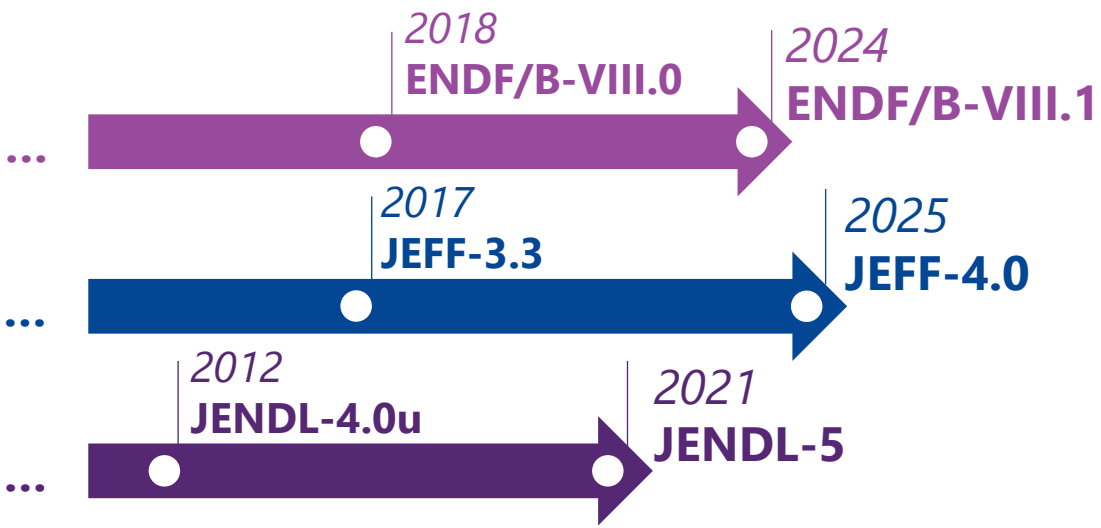
TENDL, CENDL, BROADR, …



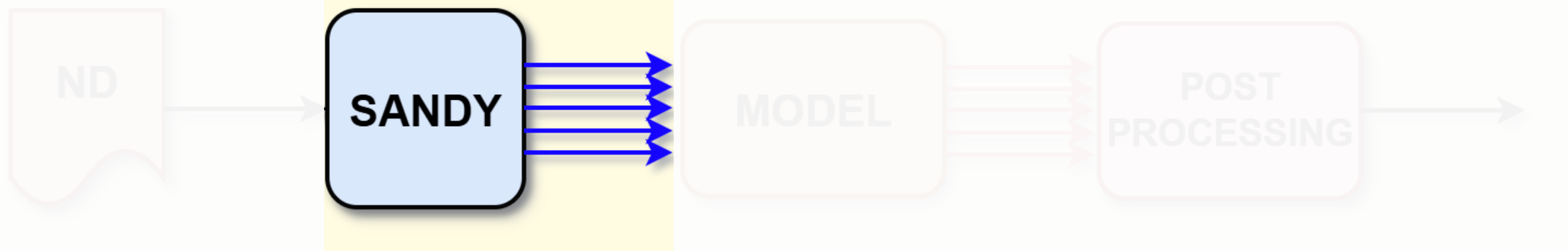
Nuclear data evaluation project



TENDL, CENDL, BROADR, …

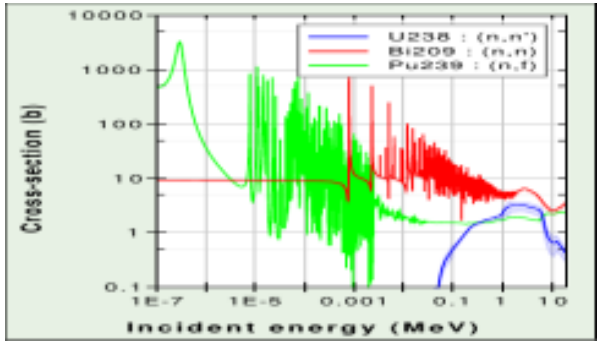


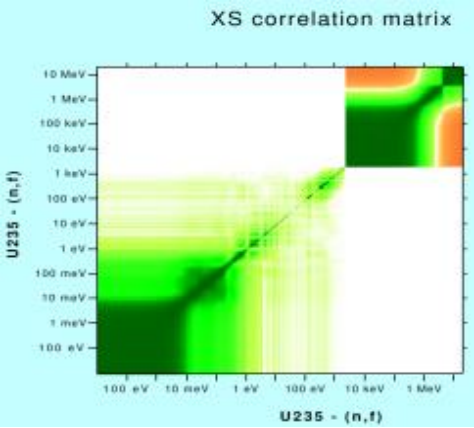
Better Nuclear data
=
Better model predictions



2. Stochastic sampling

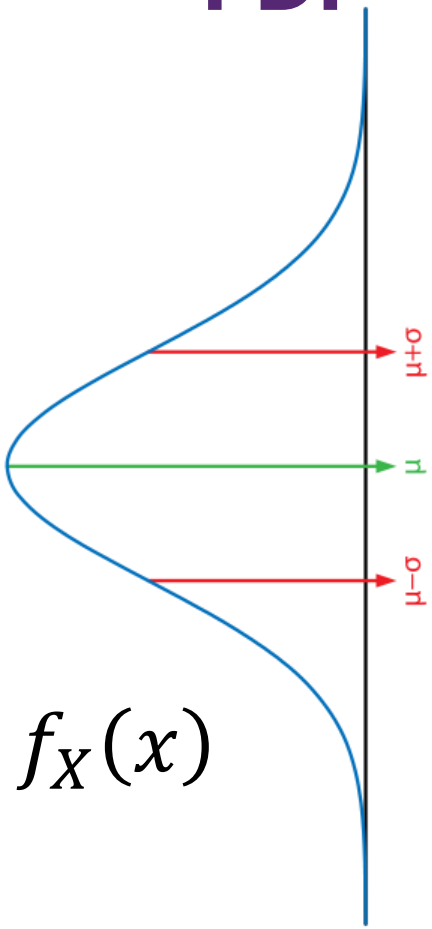
Best estimate





Uncertainty / Covariance matrix

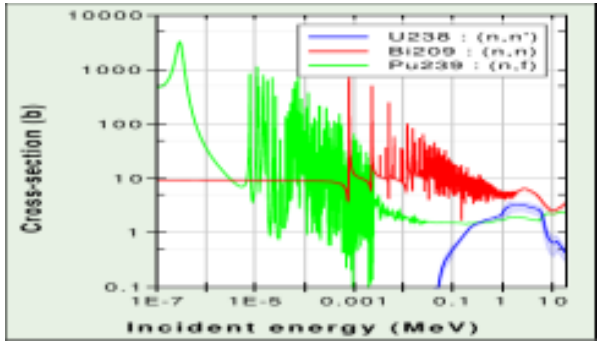
PDF

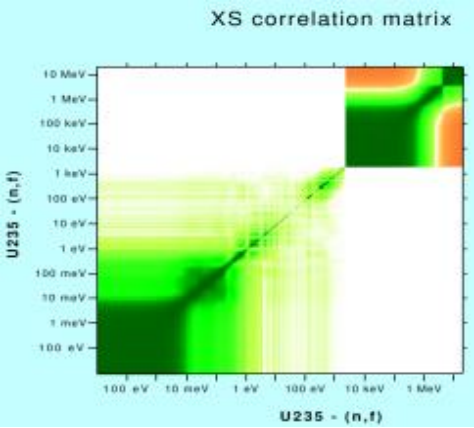


$f_X(x)$

It is widely accepted that uncertainties of physical quantities should correspond to standard deviations of event sampling probability functions (Smith, 2015)

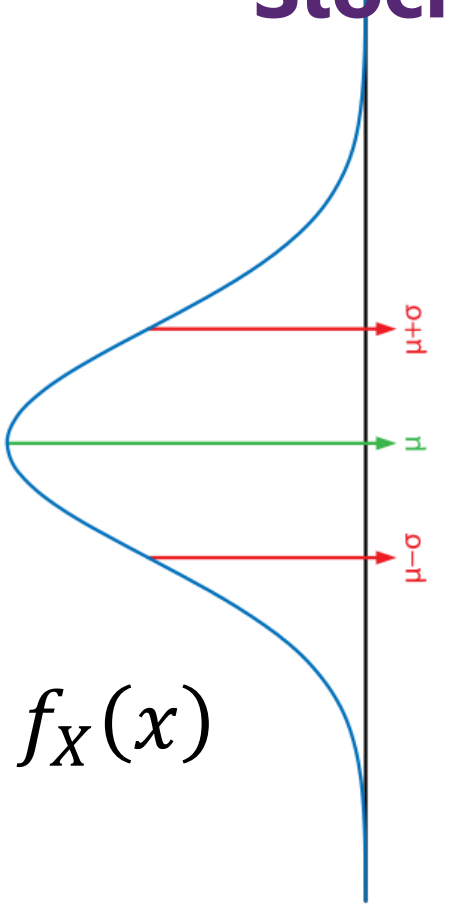
Best estimate





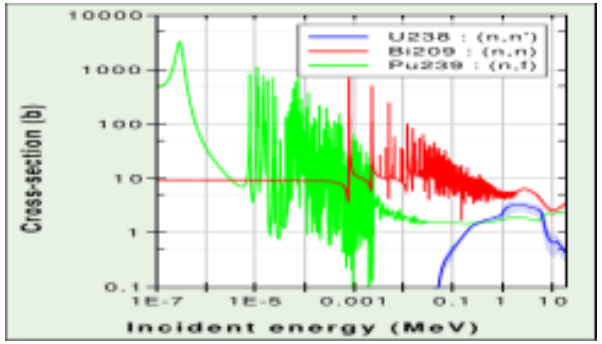
Uncertainty / Covariance matrix

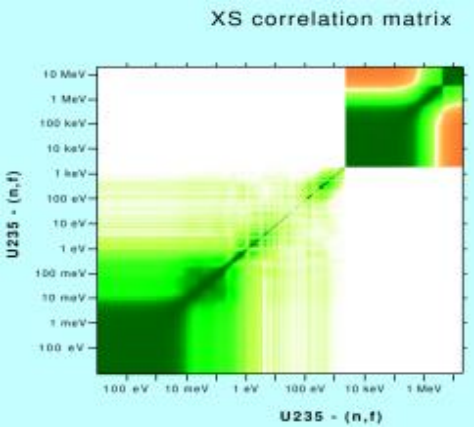
Stochastic sampling



$f_X(x)$

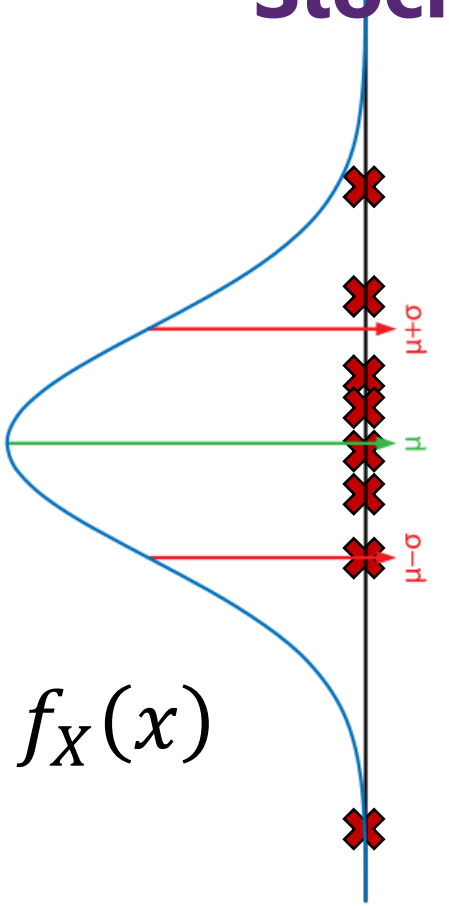
Best estimate





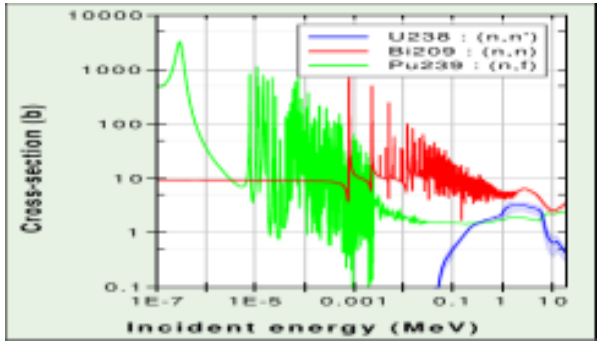
Uncertainty / Covariance matrix

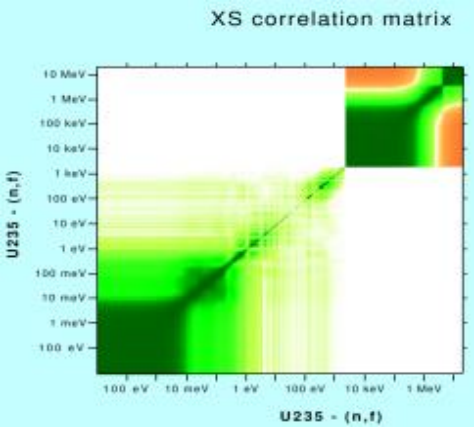
Stochastic sampling



$f_X(x)$

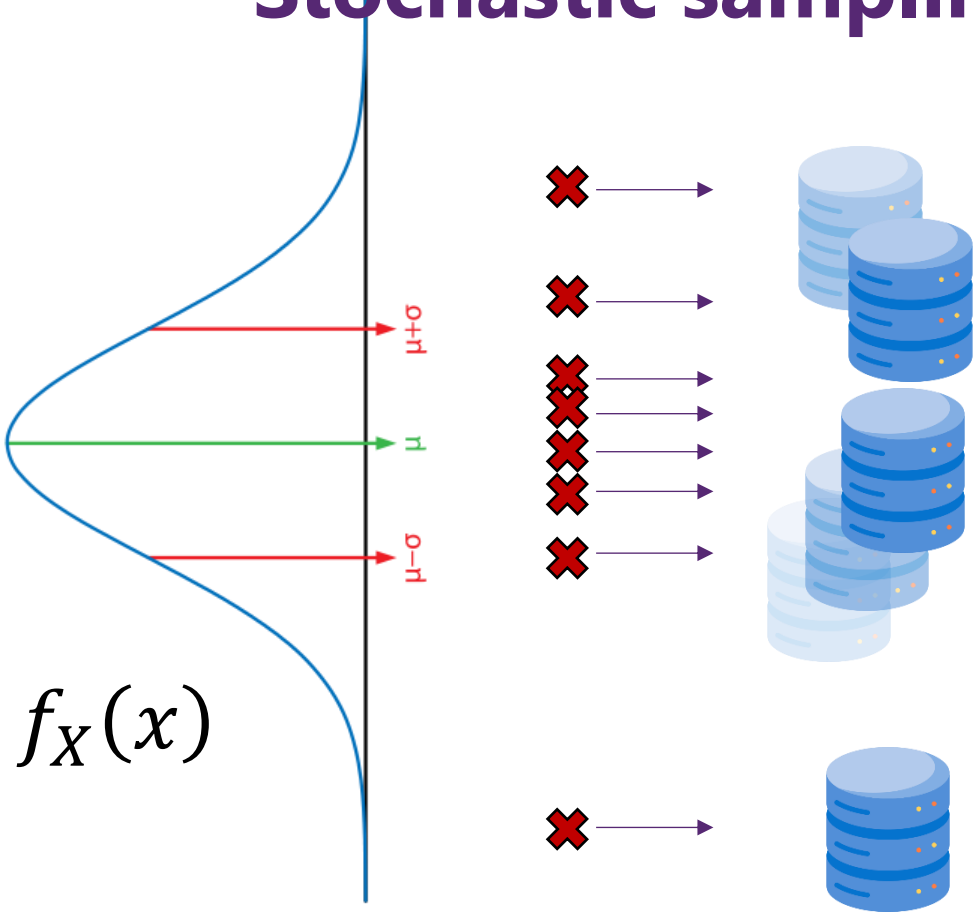
Best estimate



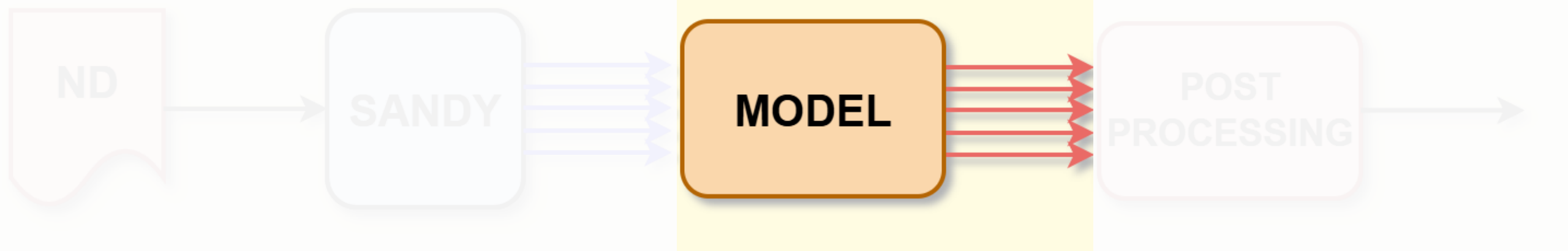


Uncertainty / Covariance matrix

Stochastic sampling



Perturbed nuclear data files



3. Uncertainty Propagation

Any model/code

Any model/code

x

ND input

y

model output

$$x^{(1)}$$

$$x^{(2)}$$

$$\vdots$$

$$x^{(i)}$$

$$\vdots$$

$$x^{(N)}$$

$$y^{(1)}$$

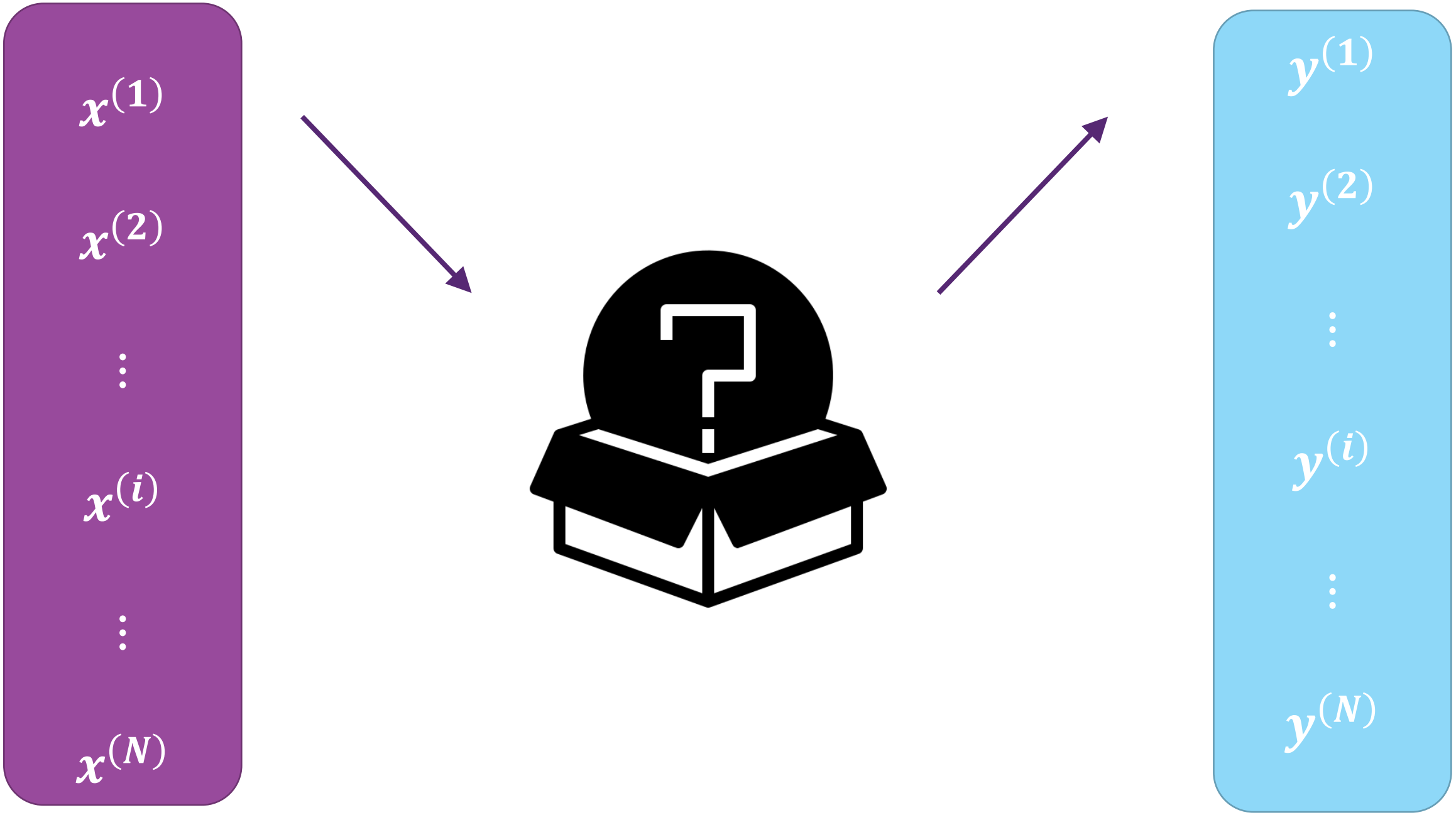
$$y^{(2)}$$

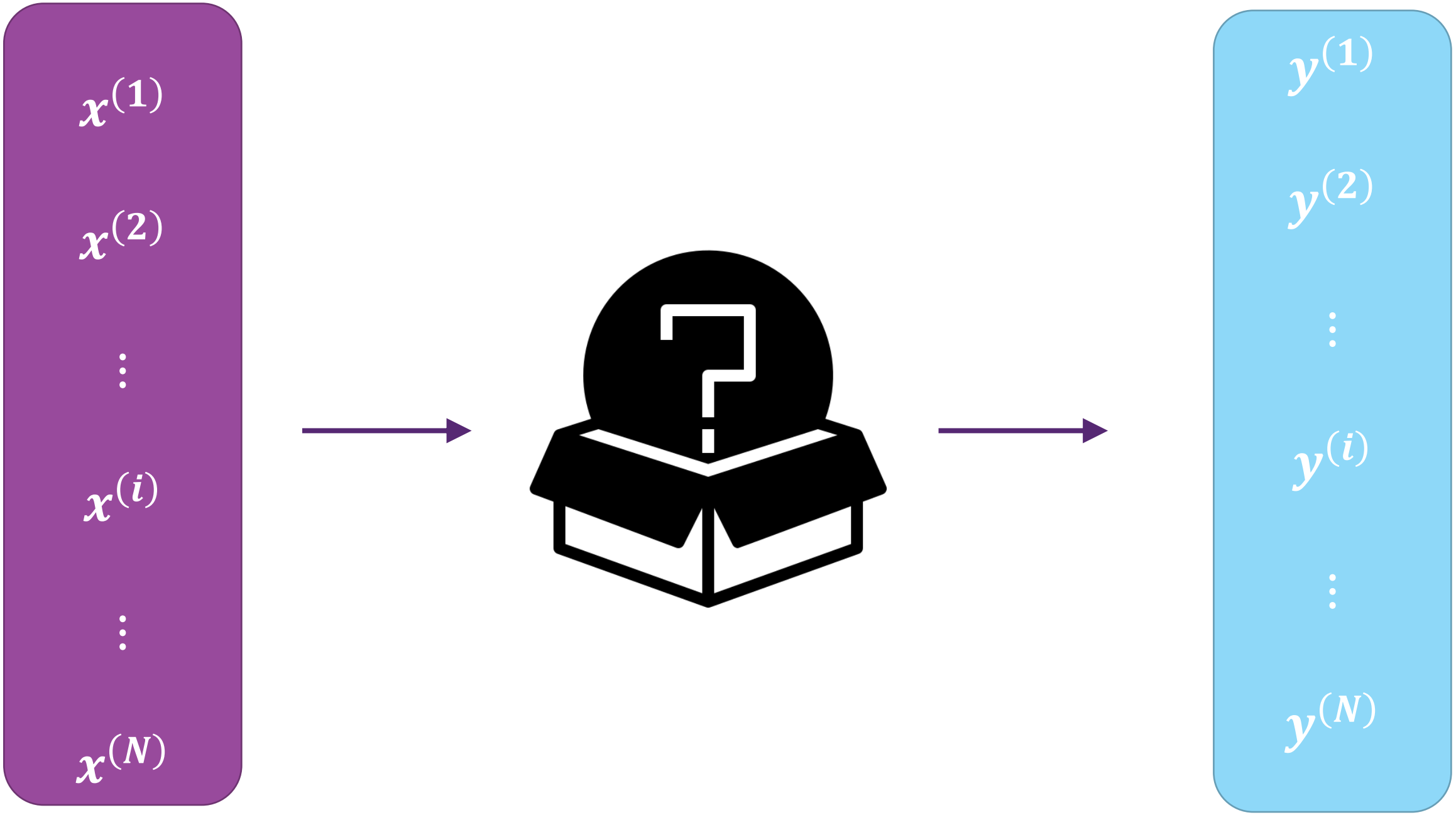
$$\vdots$$

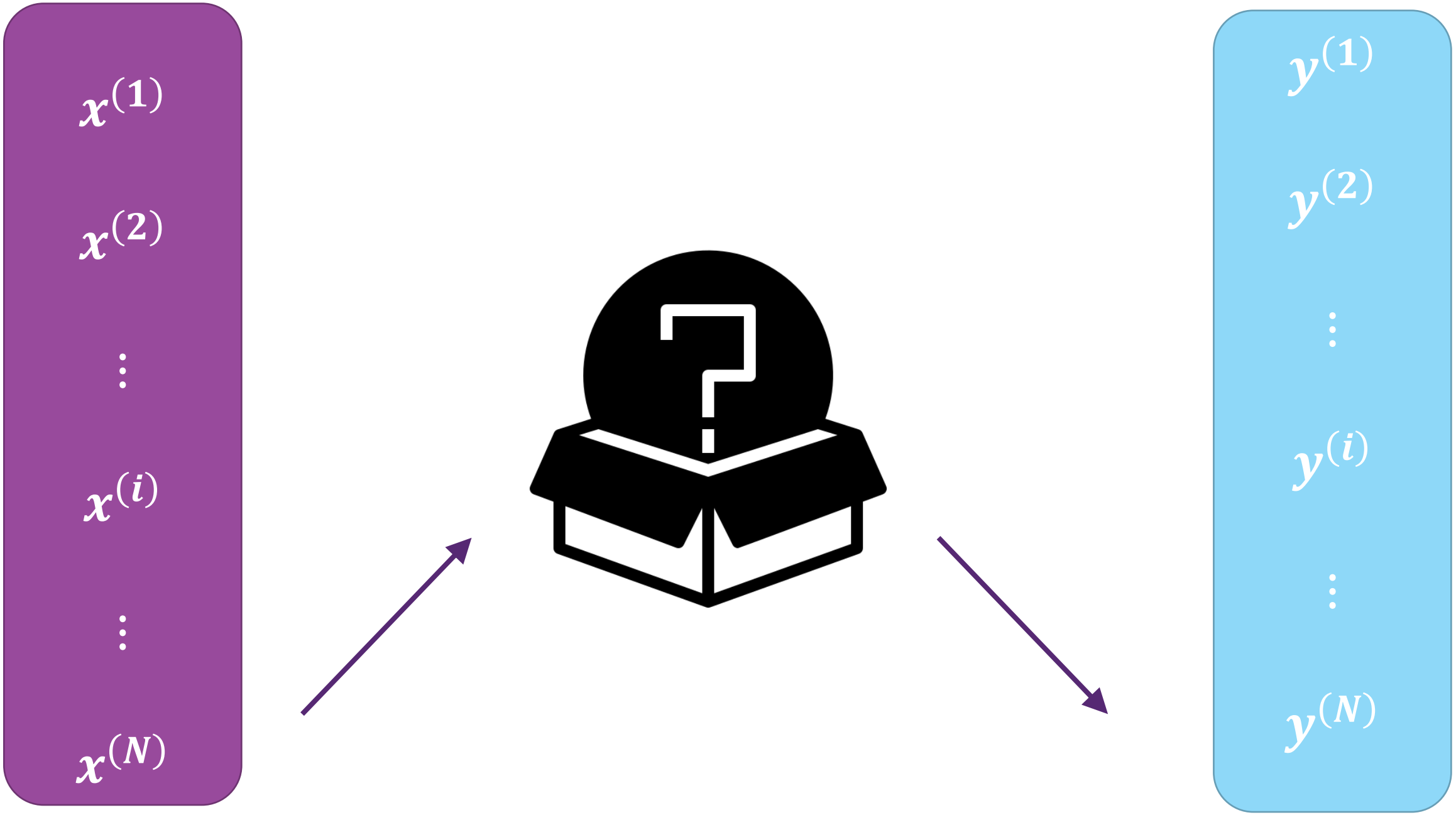
$$y^{(i)}$$

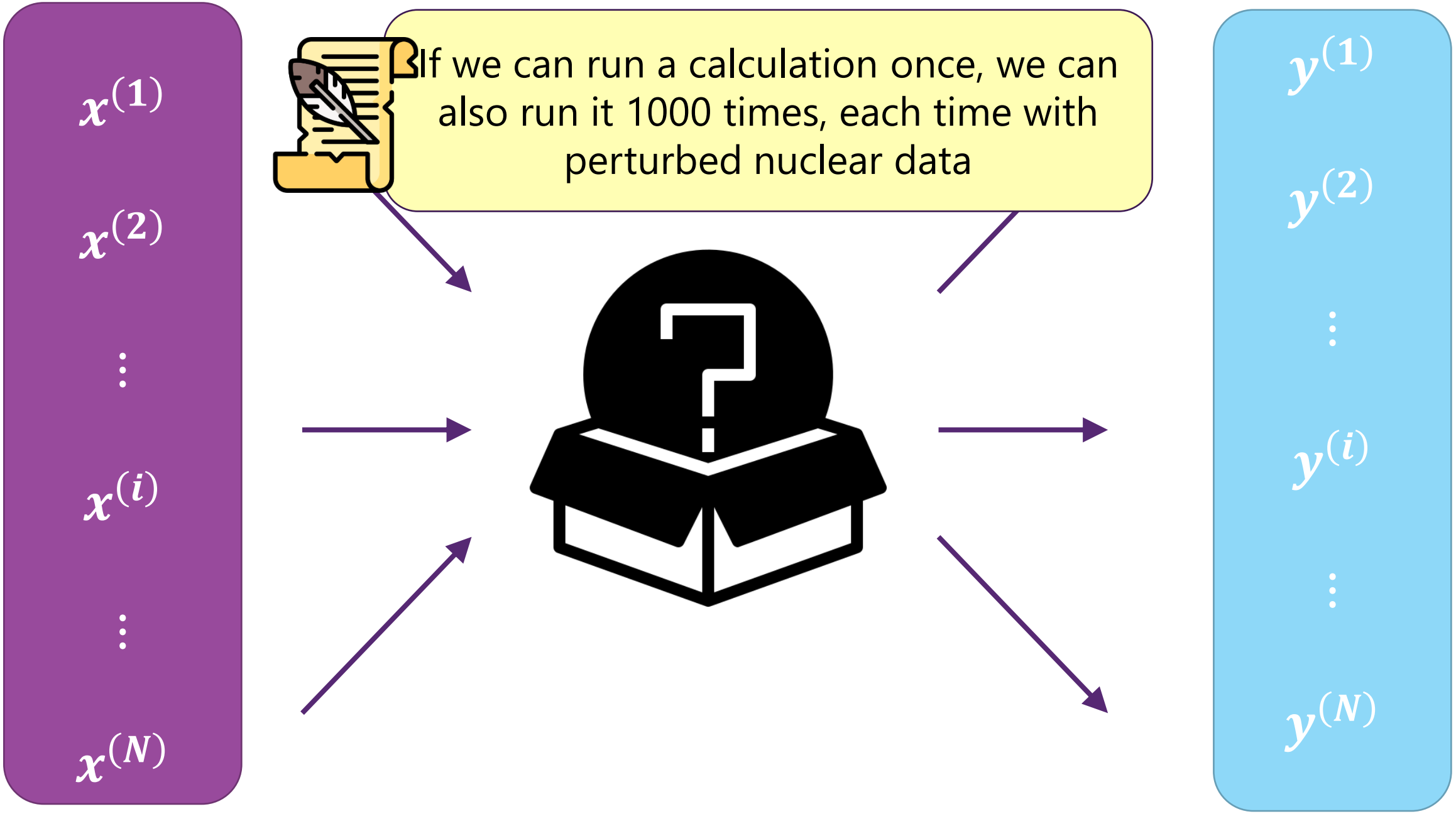
$$\vdots$$

$$y^{(N)}$$


$x^{(1)}$
$x^{(2)}$
$\vdots$
$x^{(i)}$
$\vdots$
$x^{(N)}$
$y^{(1)}$
$y^{(2)}$
$\vdots$
$y^{(i)}$
$\vdots$
$y^{(N)}$


$x^{(1)}$
$x^{(2)}$
$\vdots$
$x^{(i)}$
$\vdots$
$x^{(N)}$
$y^{(1)}$
$y^{(2)}$
$\vdots$
$y^{(i)}$
$\vdots$
$y^{(N)}$


$x^{(1)}$
$x^{(2)}$
$\vdots$
$x^{(i)}$
$\vdots$
$x^{(N)}$
$y^{(1)}$
$y^{(2)}$
$\vdots$
$y^{(i)}$
$\vdots$
$y^{(N)}$


$x^{(1)}$
$x^{(2)}$
$\vdots$
$x^{(i)}$
$\vdots$
$x^{(N)}$
If we can run a calculation once, we can also run it 1000 times, each time with perturbed nuclear data
$y^{(1)}$
$y^{(2)}$
$\vdots$
$y^{(i)}$
$\vdots$
$y^{(N)}$

$x^{(1)}$

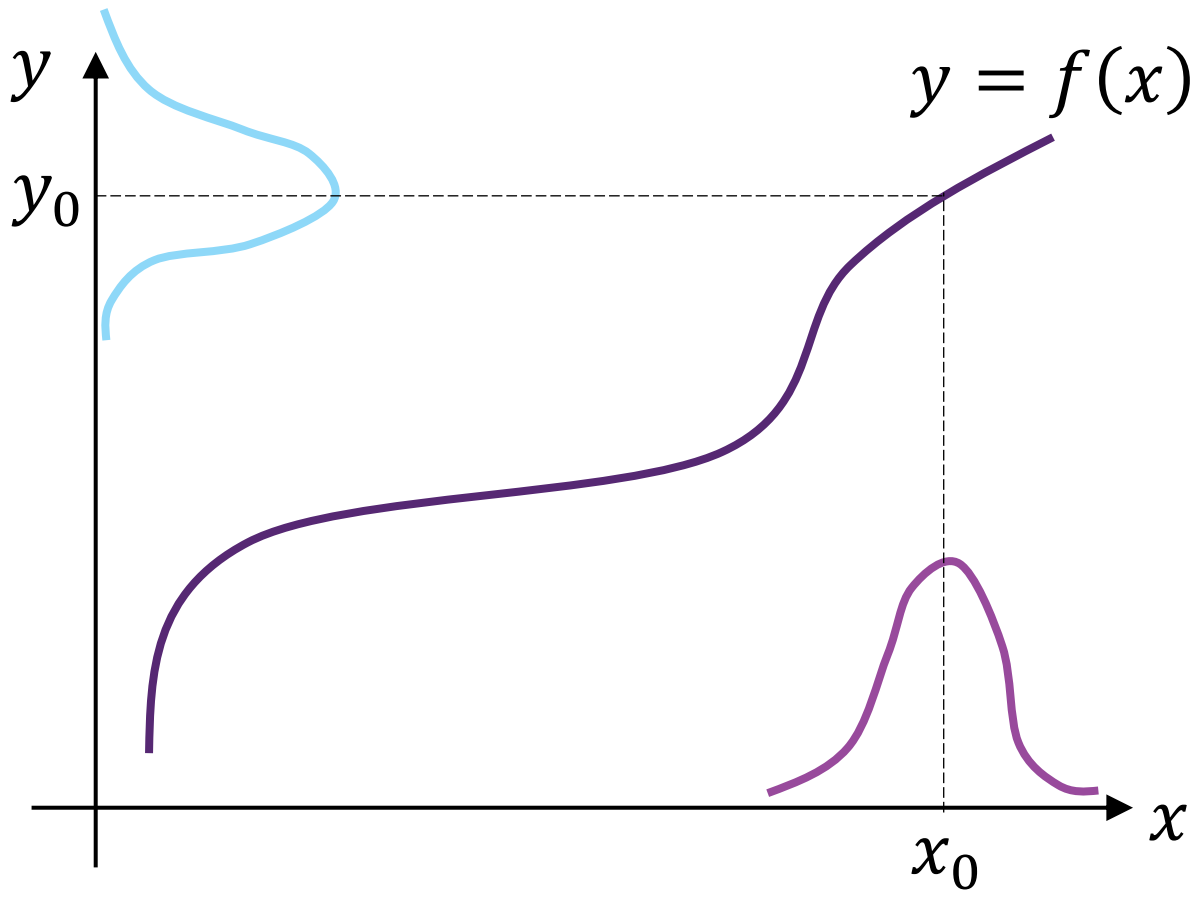
$x^{(2)}$

$\vdots$

$x^{(i)}$

$\vdots$

$x^{(N)}$



$y^{(1)}$

$y^{(2)}$

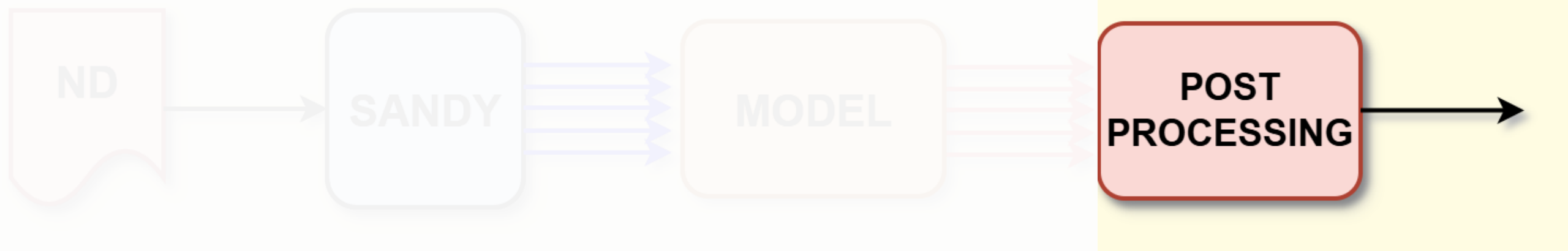
$\vdots$

$y^{(i)}$

$\vdots$

$y^{(N)}$

Convergence guaranteed for $N \to \infty$



4. Post-Processing

SAMPLE

$y^{(1)}$

$y^{(2)}$

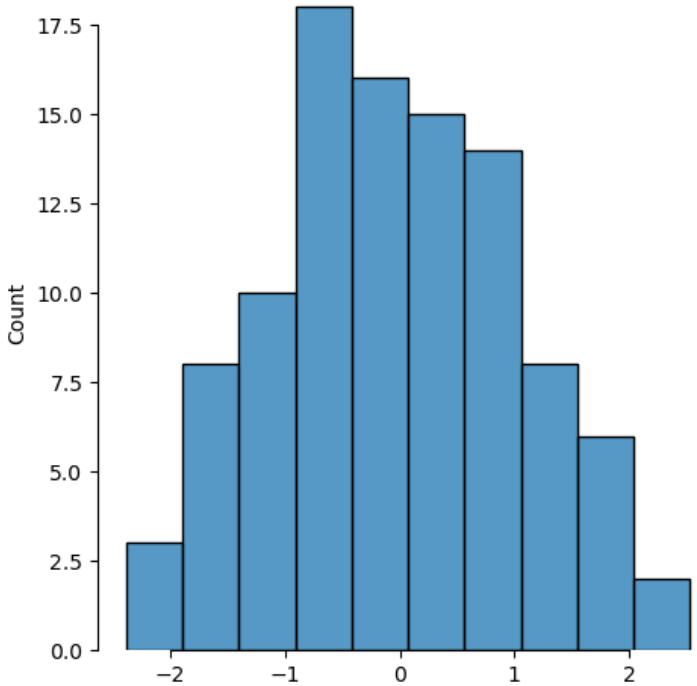
$\vdots$

$y^{(i)}$

$\vdots$

$y^{(N)}$

Statistics to estimate population parameters



- **Sample** (size N)
- **Sample mean**:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- **Sample variance**:

$$s^2 = \frac{1}{(N-1)}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

- **Variance of the sample mean**:

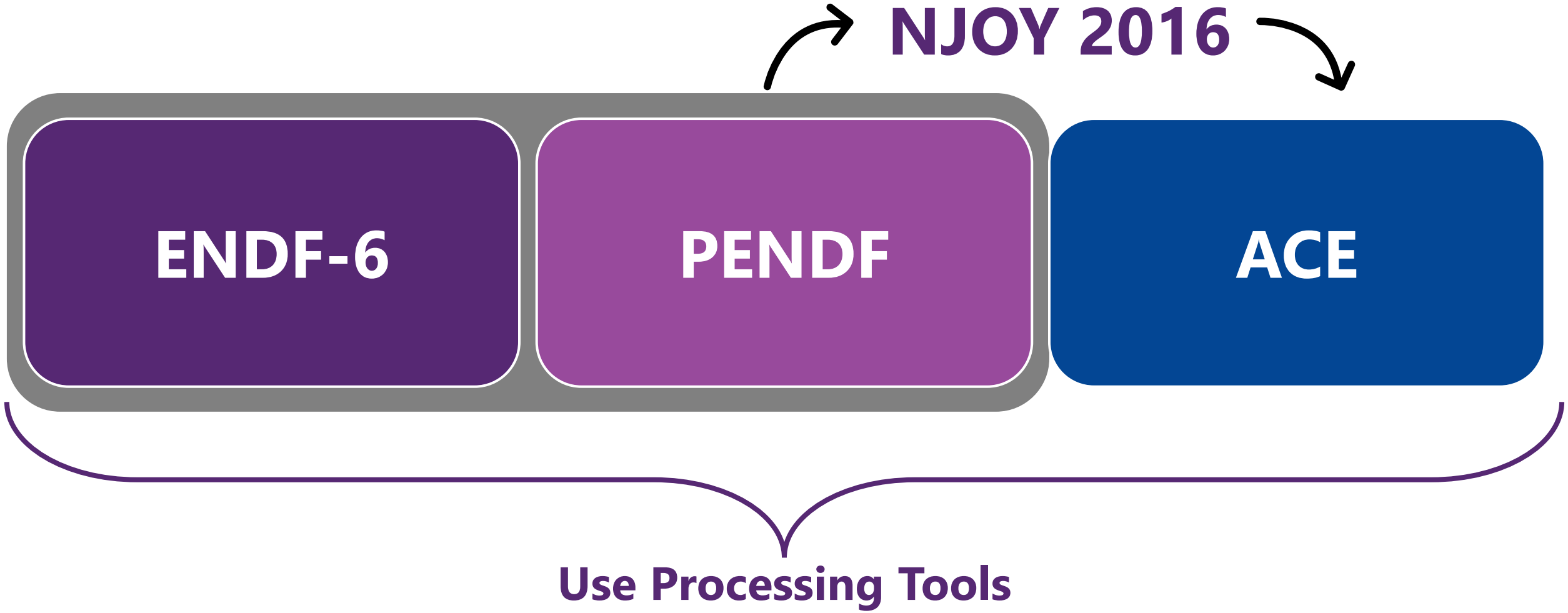
$$Var(\bar{x}) = \frac{s^2}{N}$$

ISC: Public

Q&A

?

What ND format is supported by SANDY?

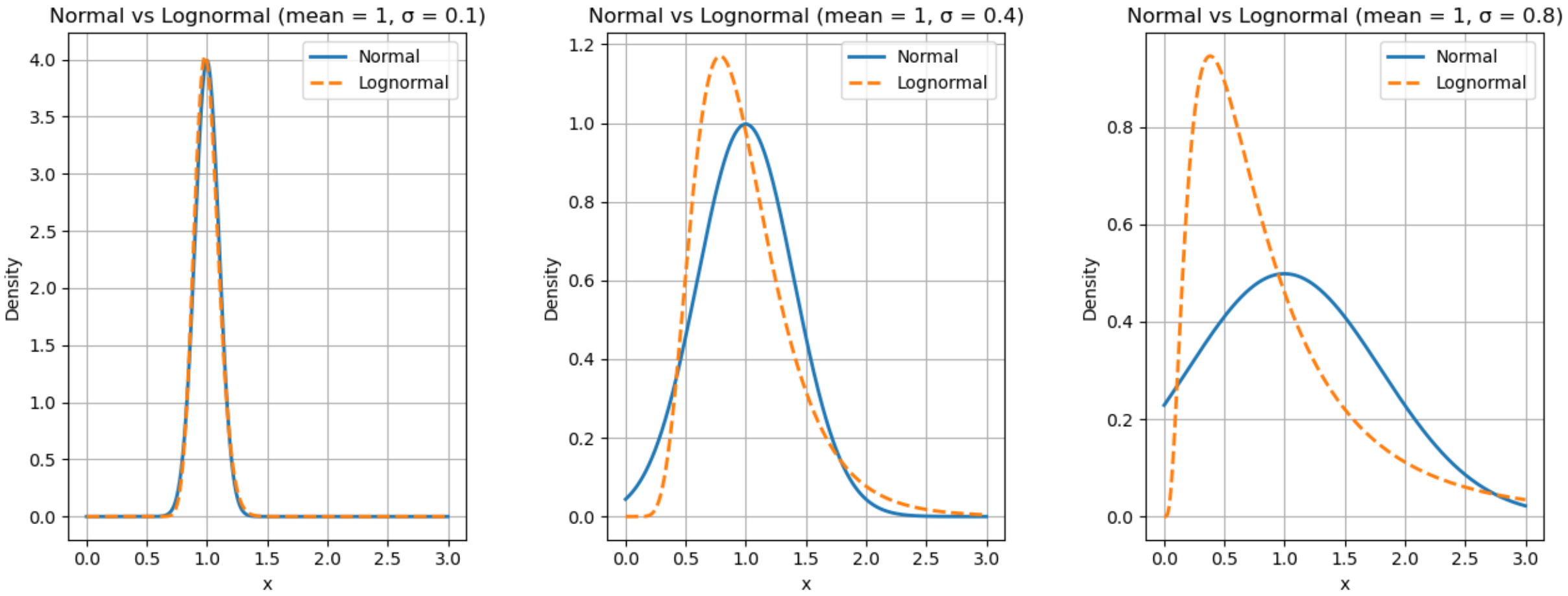


https://doi.org/10.1016/j.pnucene.2025.105892

?

What PDF should I use?

What PDF should I use?

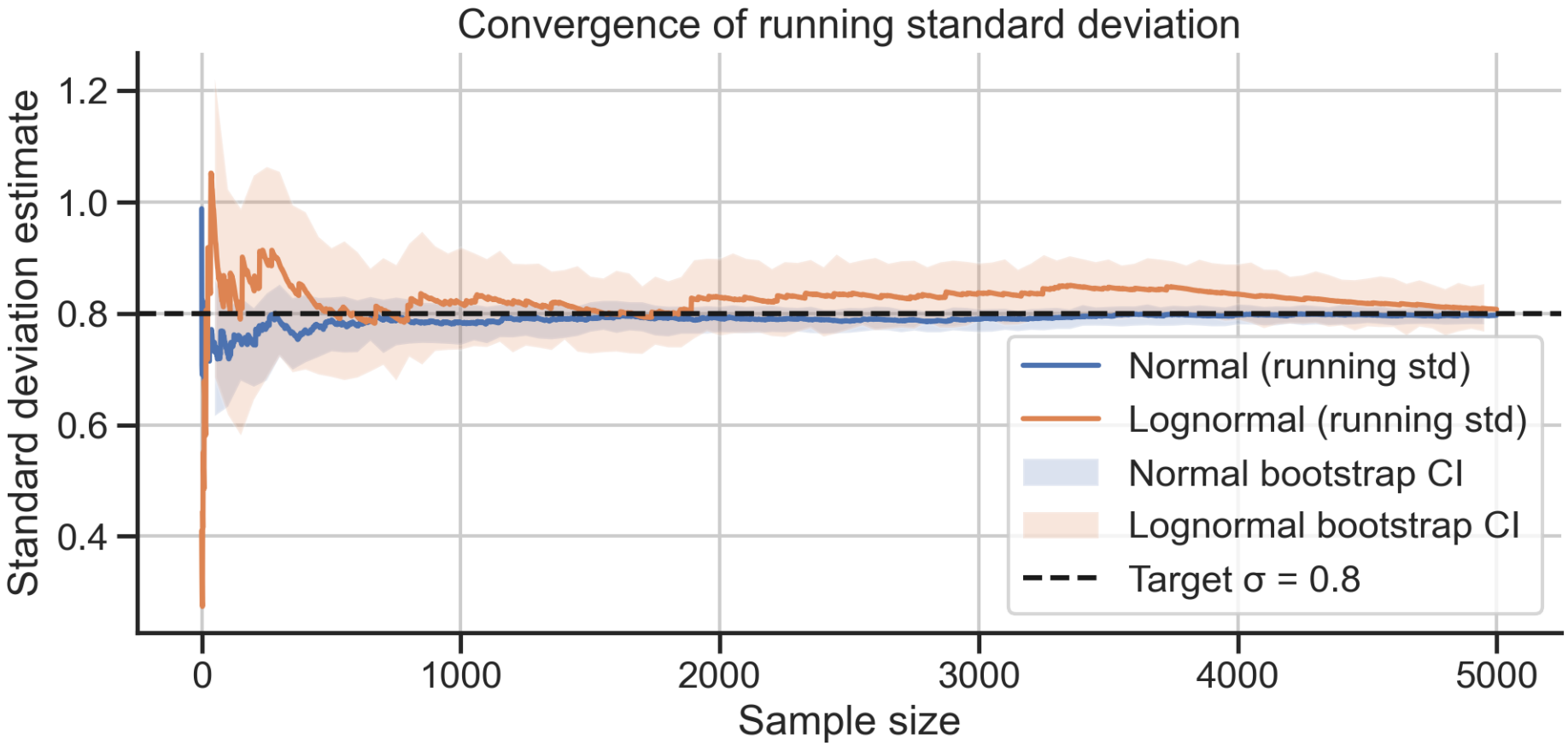


https://doi.org/10.1080/00295639.2024.2323217

?

What sample size should I select?

What sample size should I select?



https://doi.org/10.1080/00295639.2024.2353987

?

What happens if my model f is also Monte Carlo-based?

What happens if my model f is also Monte Carlo-based?

The uncertainty of the output response contains a statistical component of the MC neutron transport solver

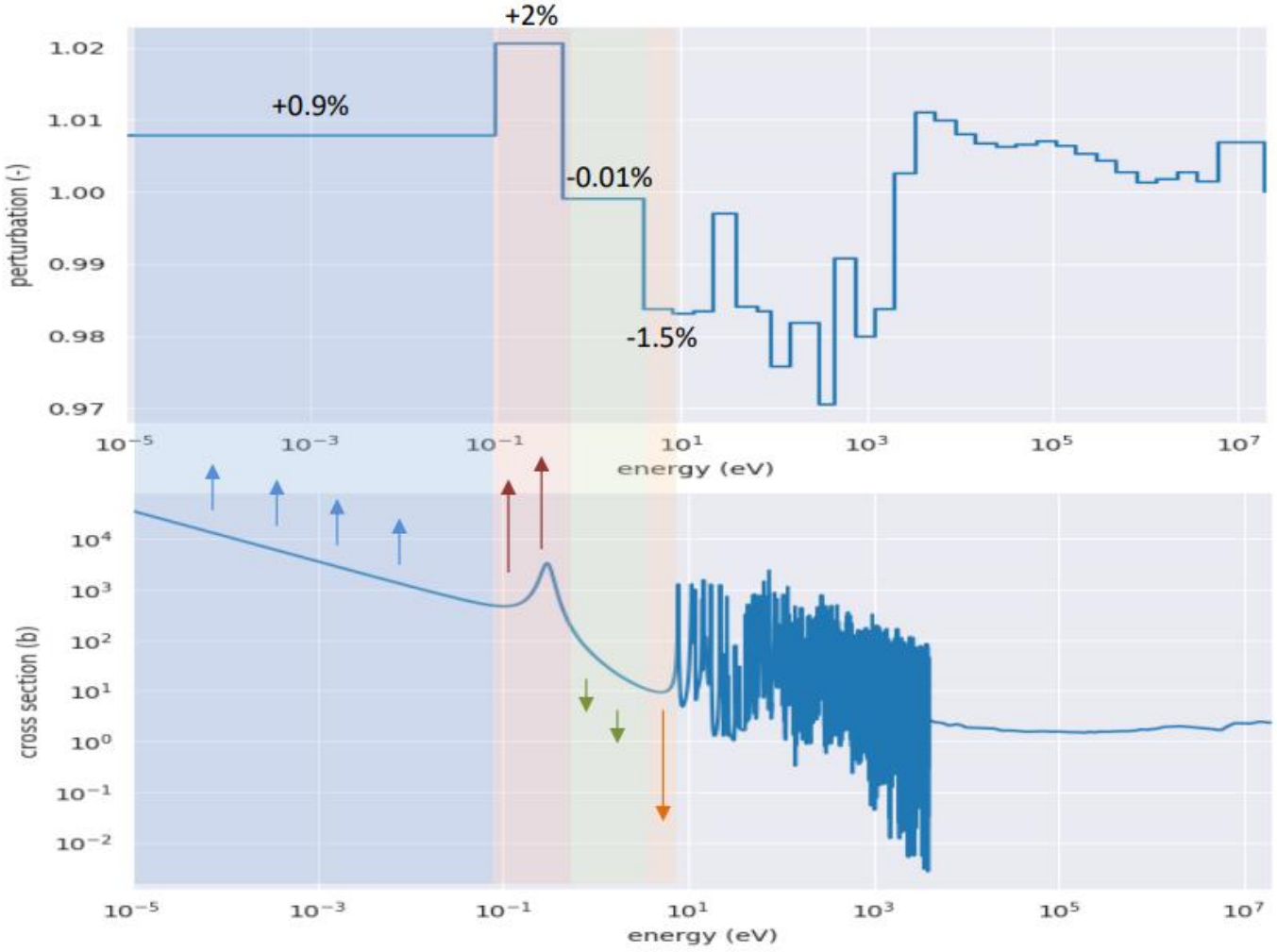
$$\sigma_{TOT}^2 = \sigma_{ND}^2 + \sigma_{MC}^2$$

https://doi.org/10.1016/j.anucene.2025.111423

?

How can MG uncertainties be transferred to CE data

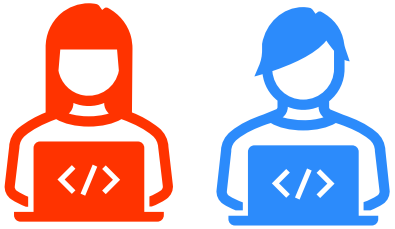
How can MG uncertainties be transferred to CE data?



SANDY workshop

1. Introduction / presentation

2. Hands-on training
   - Installation
   - Exercises



https://github.com/luca-fiorito-11/sandy

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